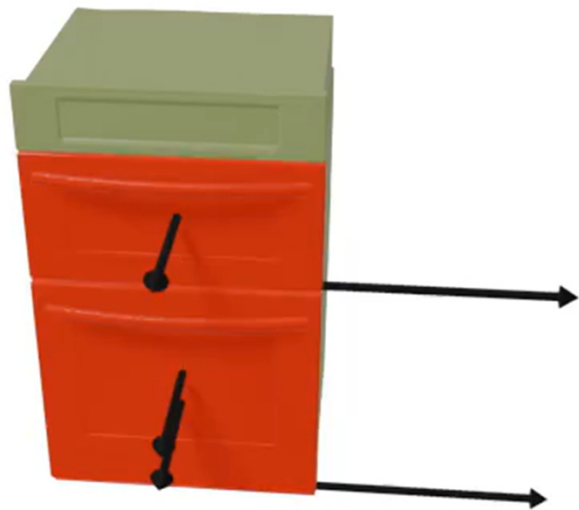
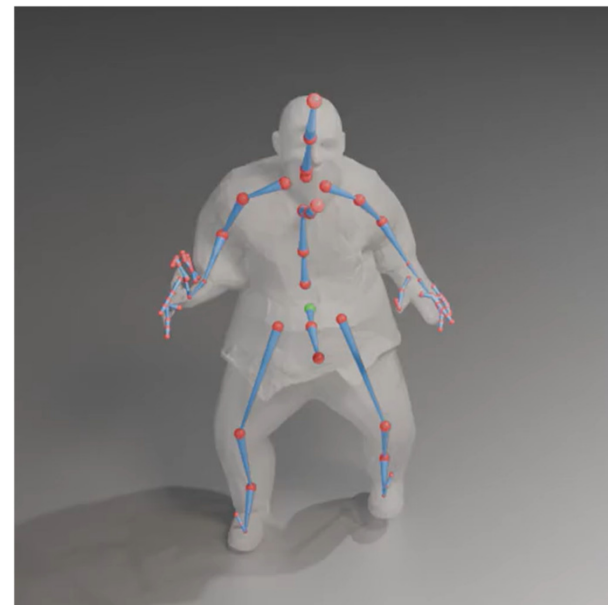


# Bringing 3D Assets to Life

Generating Articulation, Physical Properties, and Motion



Articulation



Reusable rigs



Physical response



Observed motion

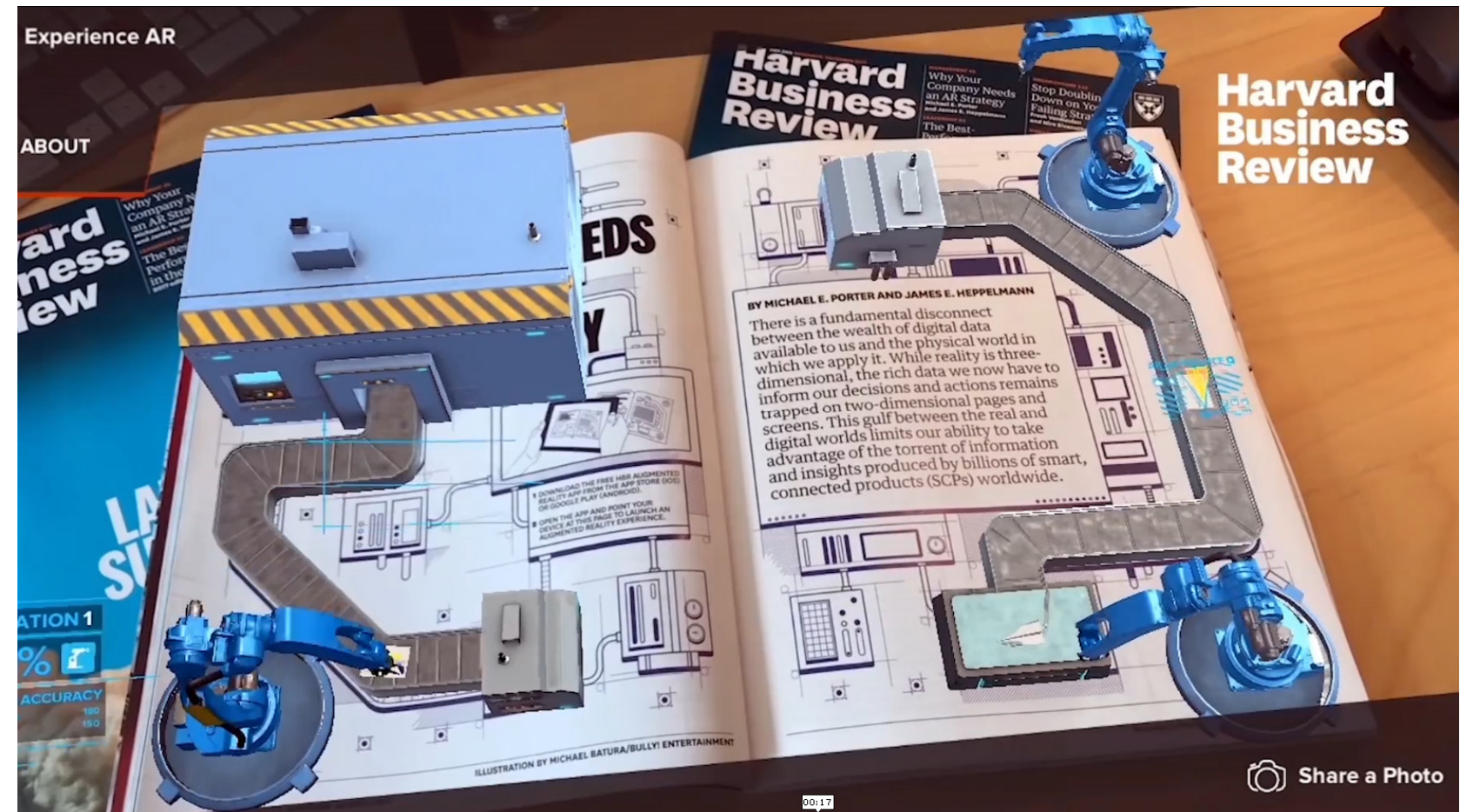
**Evangelos Kalogerakis**

TU Crete | CYENS | UMass Amherst

# Living in the 3D Digital Era!



**The Real Significance Of AR, VR And Mixed Reality** [\[forbes.com\]](https://forbes.com)



**Why Every Organization Needs an Augmented Reality Strategy** [\[hbr.org\]](https://hbr.org)

# 3D Interactive Environments



A Simulated Part-based Interactive Environment [<https://sapien.ucsd.edu/>]



RoboDK - Guide to 3D Simulation Software [[robodk.com](http://robodk.com)]

# 3D Interactive Social Media & Entertainment



Entering The Metaverse [[popularmechanics.com](https://www.popularmechanics.com)]

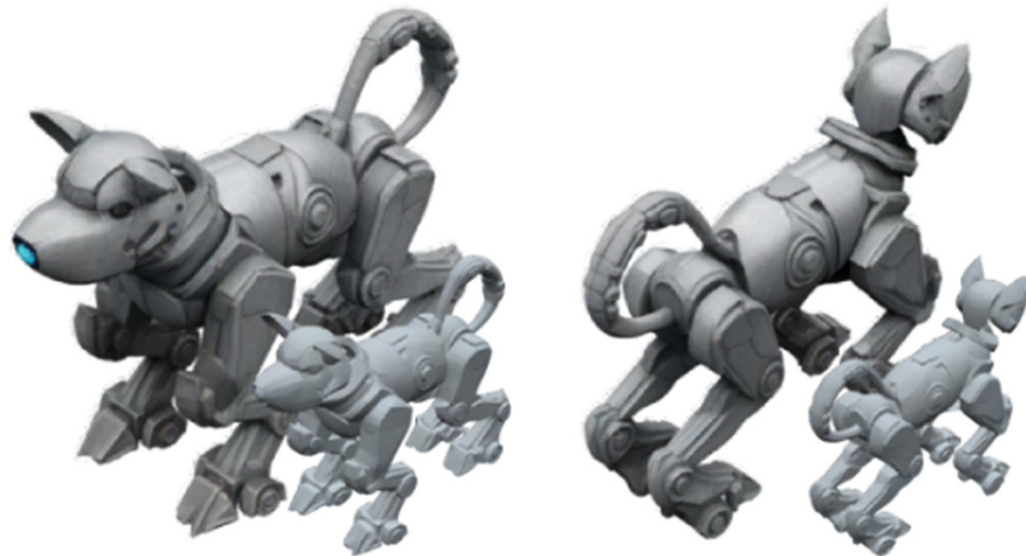


Animated Meerkat [[Weta Digital](#) / [Epic Games](#)]

# 3D Generative AI has made huge progress



*Portable transistor radio, dark cover, speaker grille, brand logo on front.*

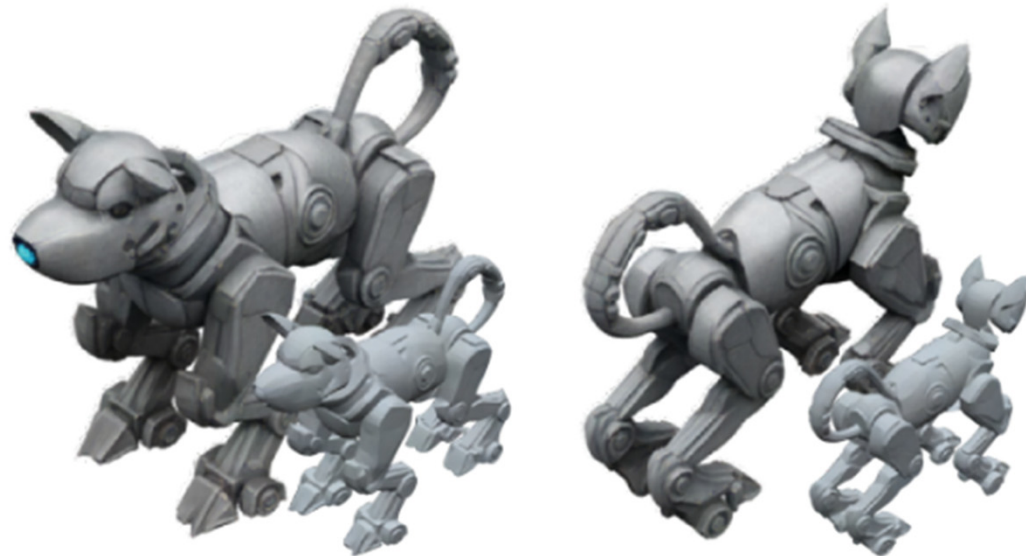


*Metallic dog-like robot with articulated legs and futuristic design elements.*

# 3D Generative AI has made huge progress



Portable transistor radio, dark cover, speaker grille, brand logo on front.



Metallic dog-like robot with articulated legs and futuristic design elements.

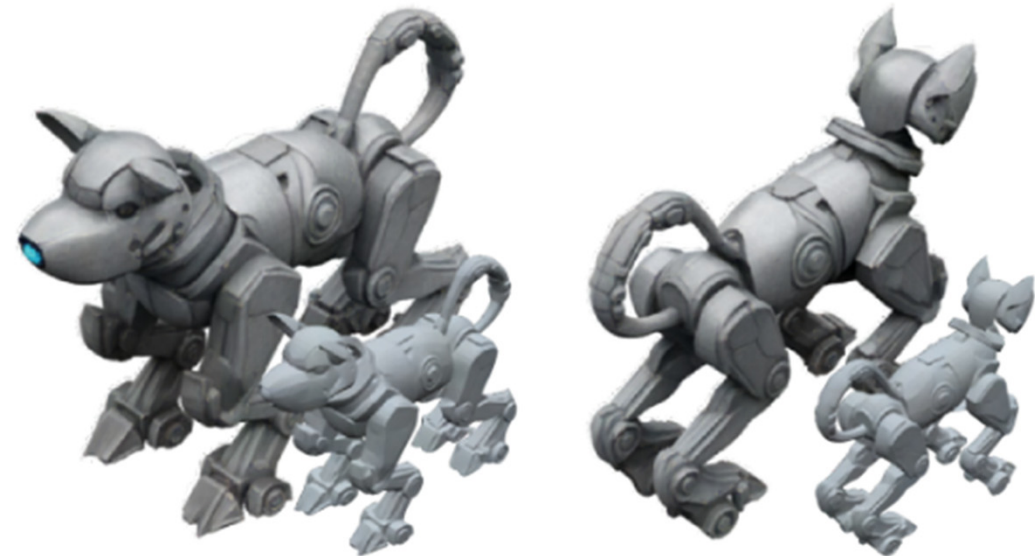


[TRELLIS, CVPR 2025]

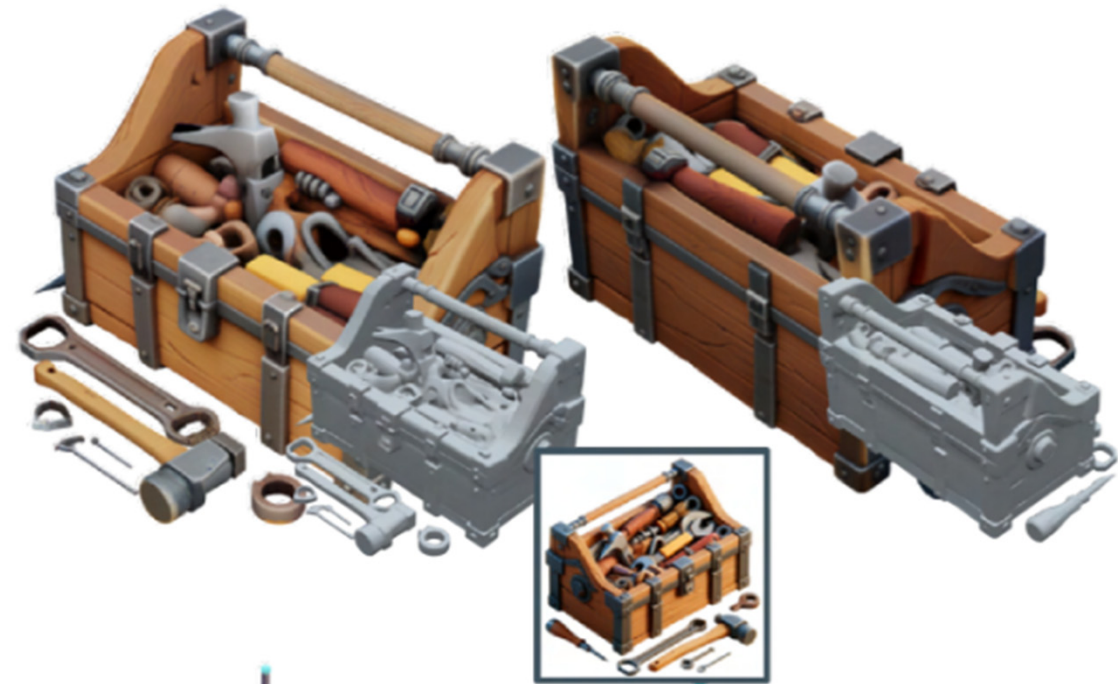
# 3D Generative AI has made huge progress



Portable transistor radio, dark cover, speaker grille, brand logo on front.



Metallic dog-like robot with articulated legs and futuristic design elements.



[TRELLIS, CVPR 2025]

However...

These are “DEAD” objects  
(static, non-interactive)



Portable transistor radio, dark cover, speaker grille, brand logo on front



Metallic dog-like robot with articulated legs and futuristic design elements.



# A convincing surface is only the beginning...

What must an object represent before we can interact with it?



**OPEN IT**

Which parts can move?



**POSE IT**

Which controls deform it?



**DROP IT**

How does it respond?

# Roadmap

1

**GEOPARD**

What articulations  
are plausible for  
given assets?

2

**Rigel3D**

How to generate  
assets with their  
articulation?

3

**SOPHY**

How should assets  
respond in a  
physical world?

4

**BLARM**

How can we  
drive the motion  
of assets?

# Roadmap

1

## GEOPARD

What articulations are plausible for given assets?

2

## Rigel3D

How to generate assets with their articulation?

3

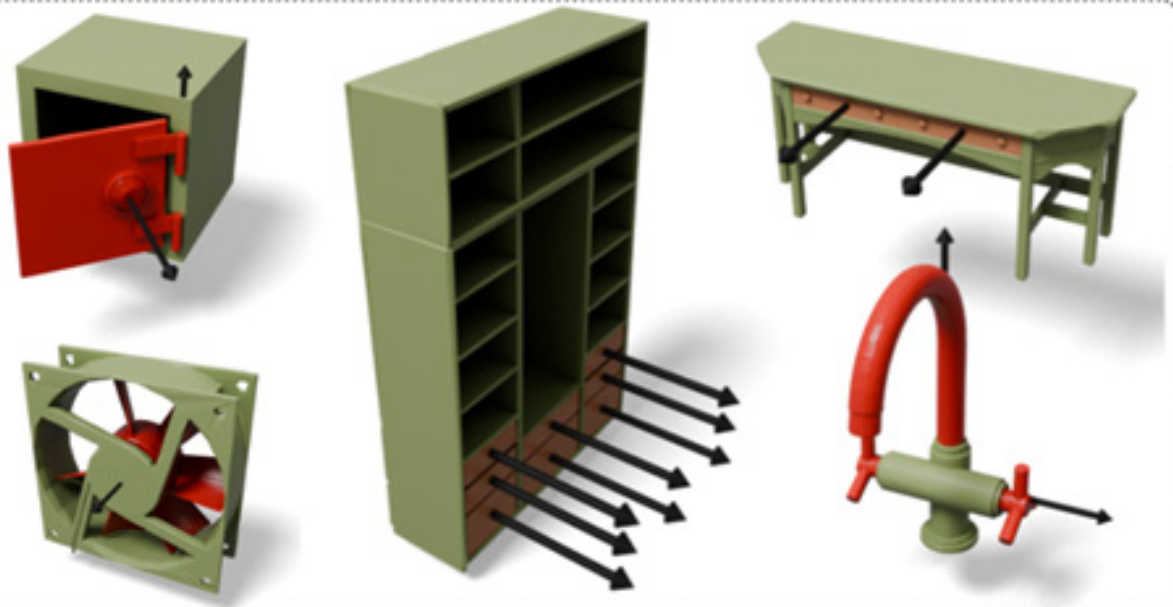
## SOPHY

How should assets respond in a physical world?

4

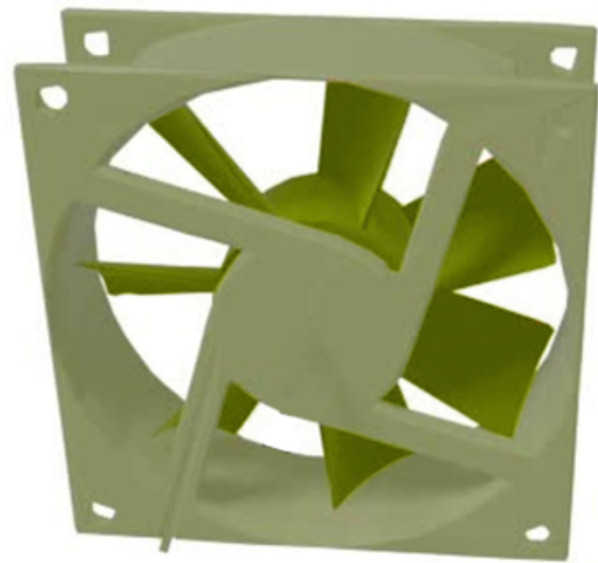
## BLARM

How can we drive the motion of assets?

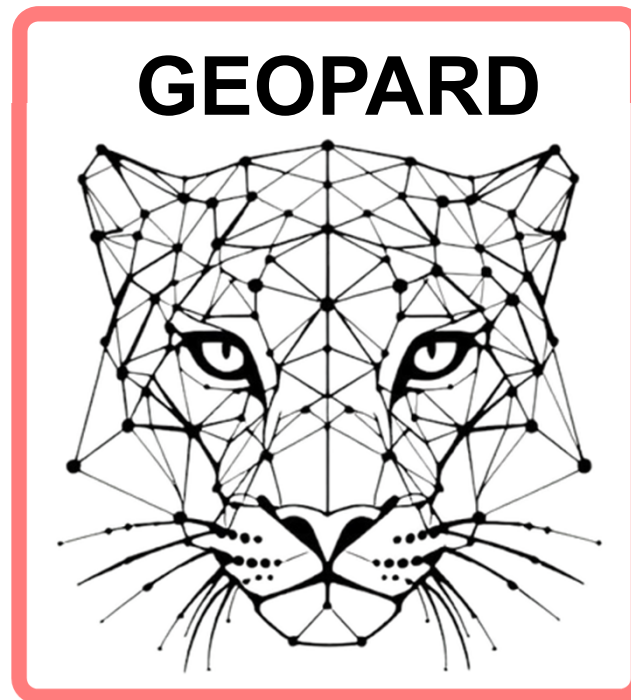
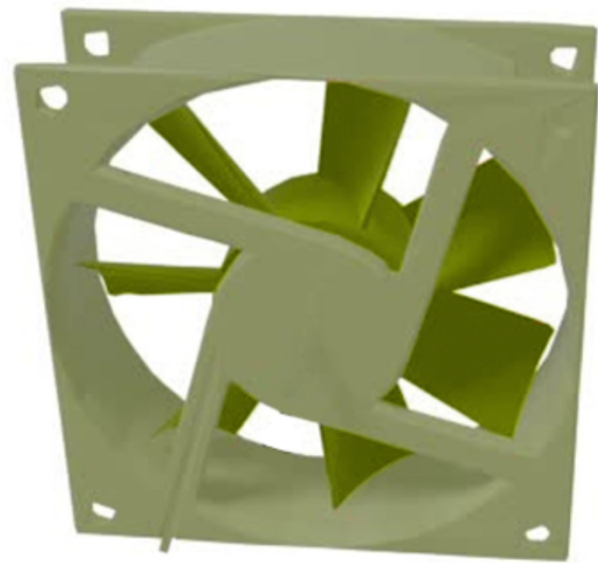


Pradyumn Goyal, D. Petrov, S. Andrews, Y.B. Shabat, H.T. Derek Liu, E. Kalogerakis, ICCV 25  
**GEOPARD**: Geometric Pretraining for Articulation Prediction in 3D Shapes  
<https://prady1272.github.io/GEOPARD/>

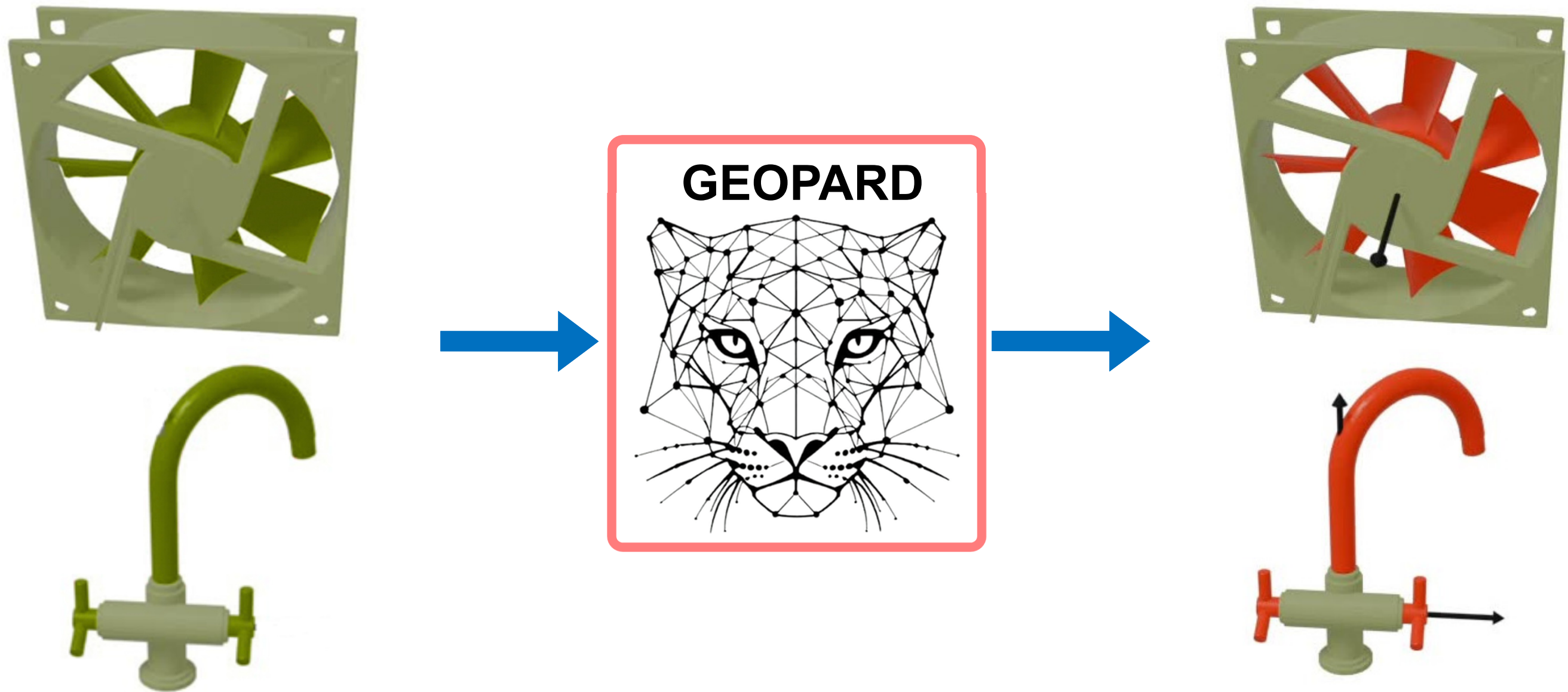
Goal: estimate articulation from a static asset



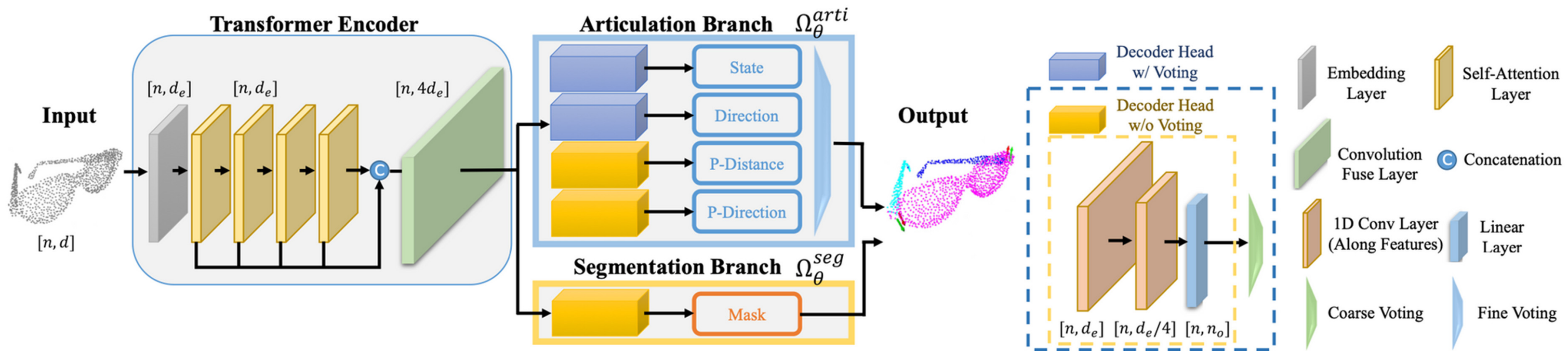
Goal: estimate articulation from a static asset



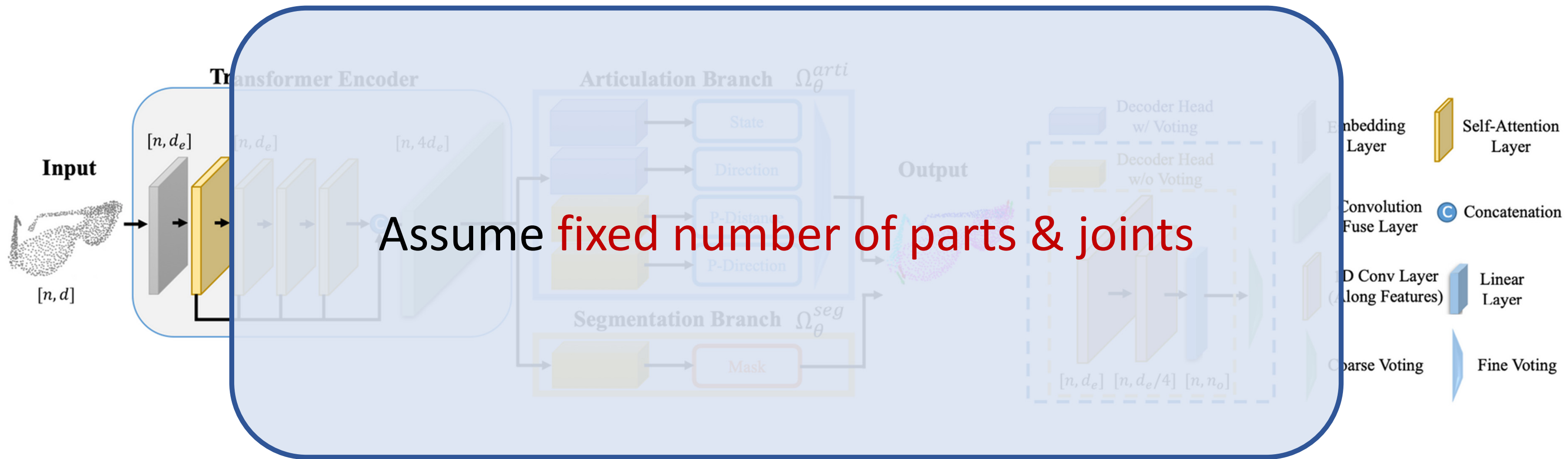
Goal: estimate articulation from a static asset



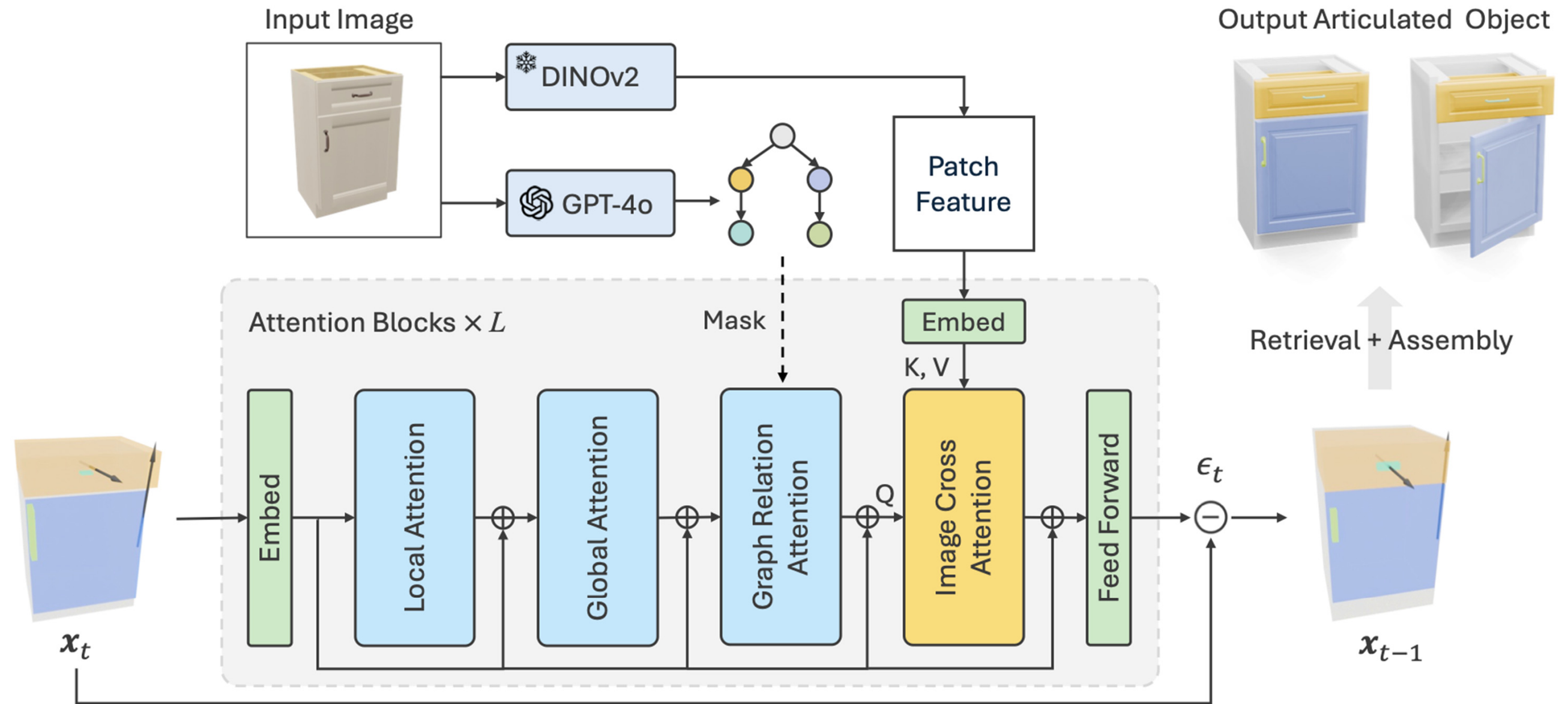
# Prior Work



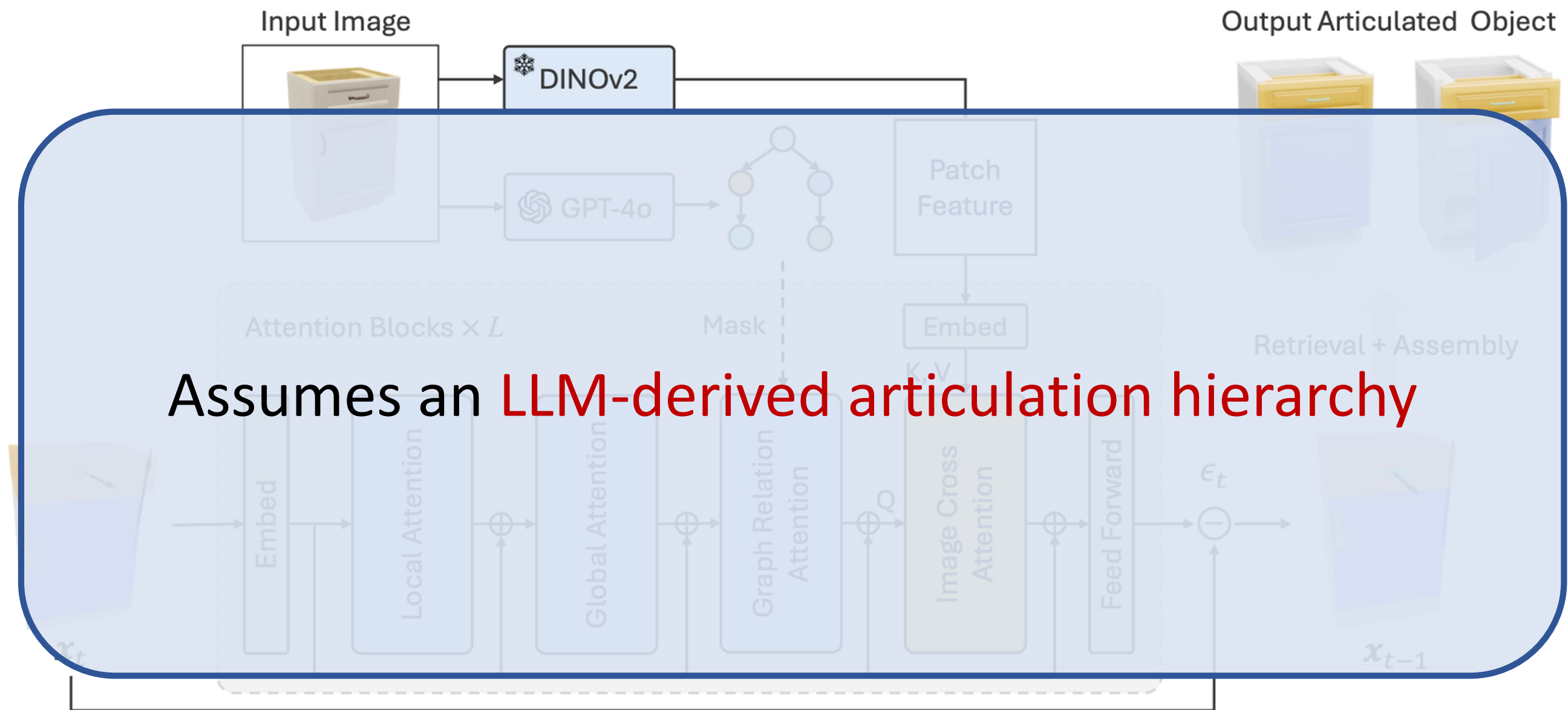
# Prior Work



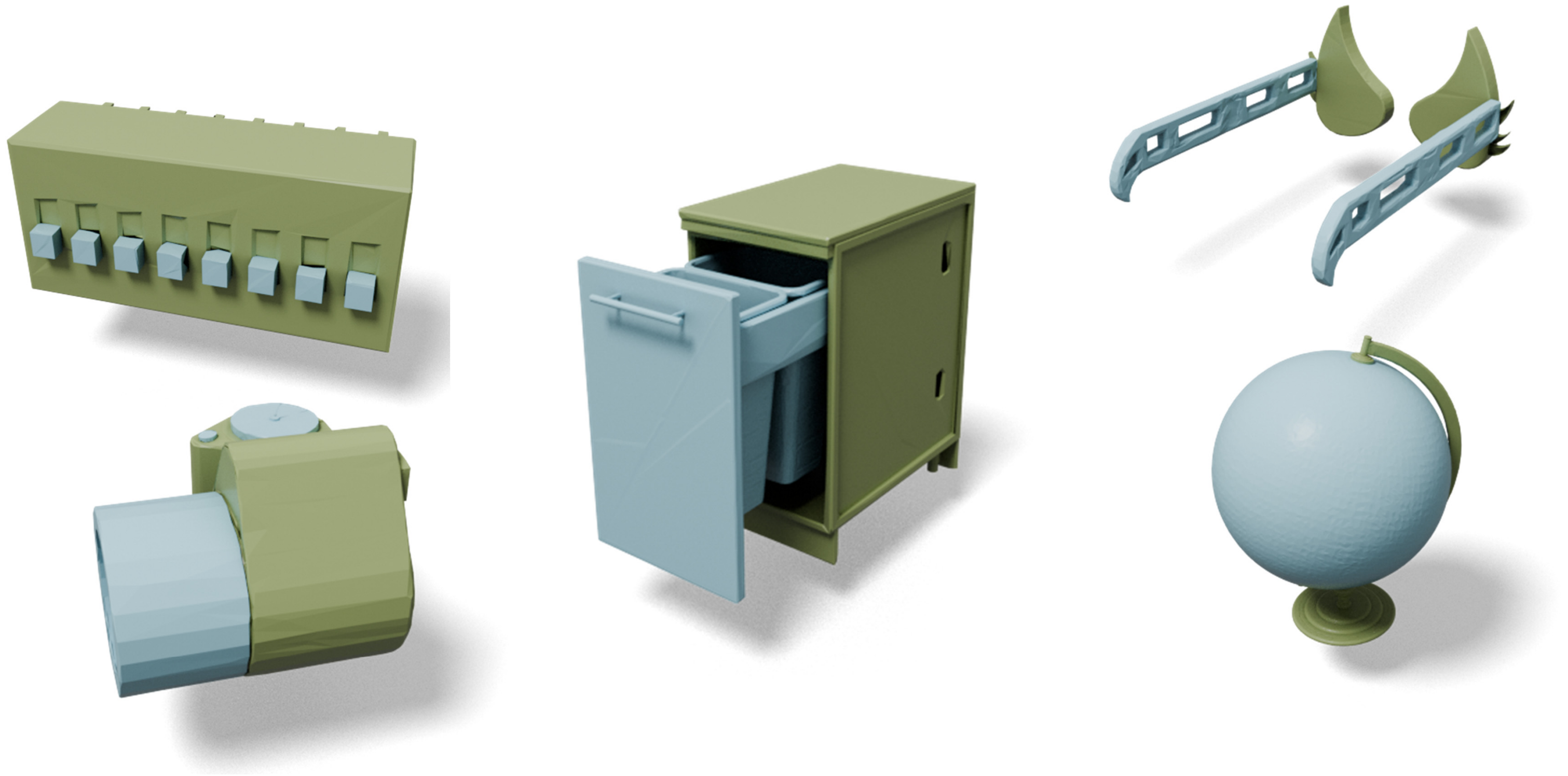
# Prior Work



# Prior Work



Input assumption: 3D segmented models



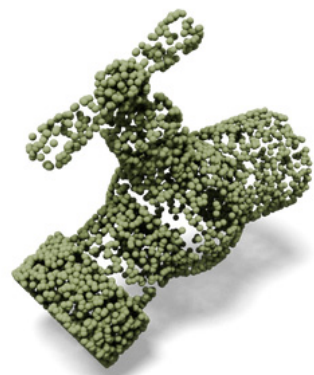
# A part makes sense in the context of its object



PART



Part tokens

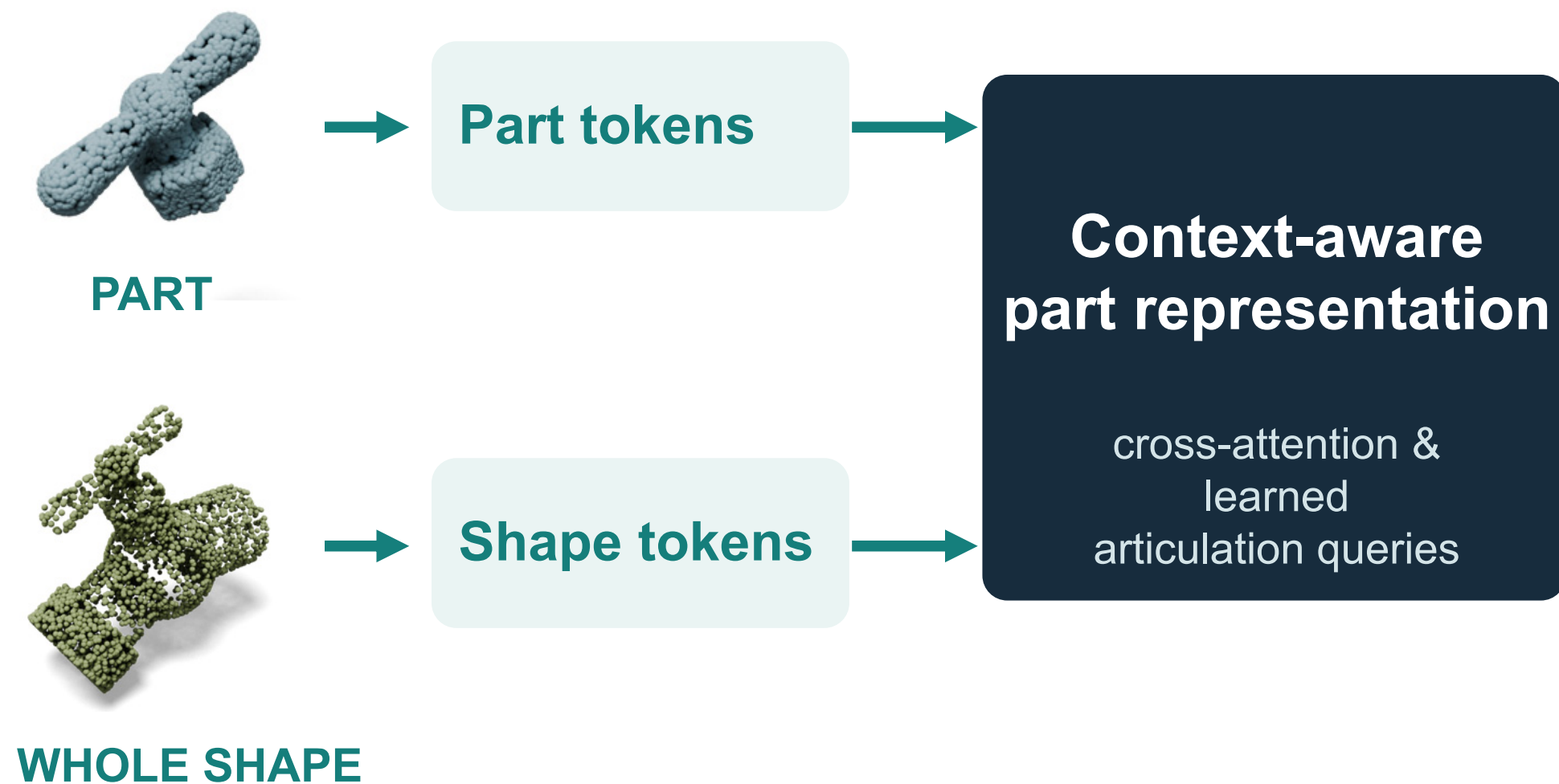


WHOLE SHAPE

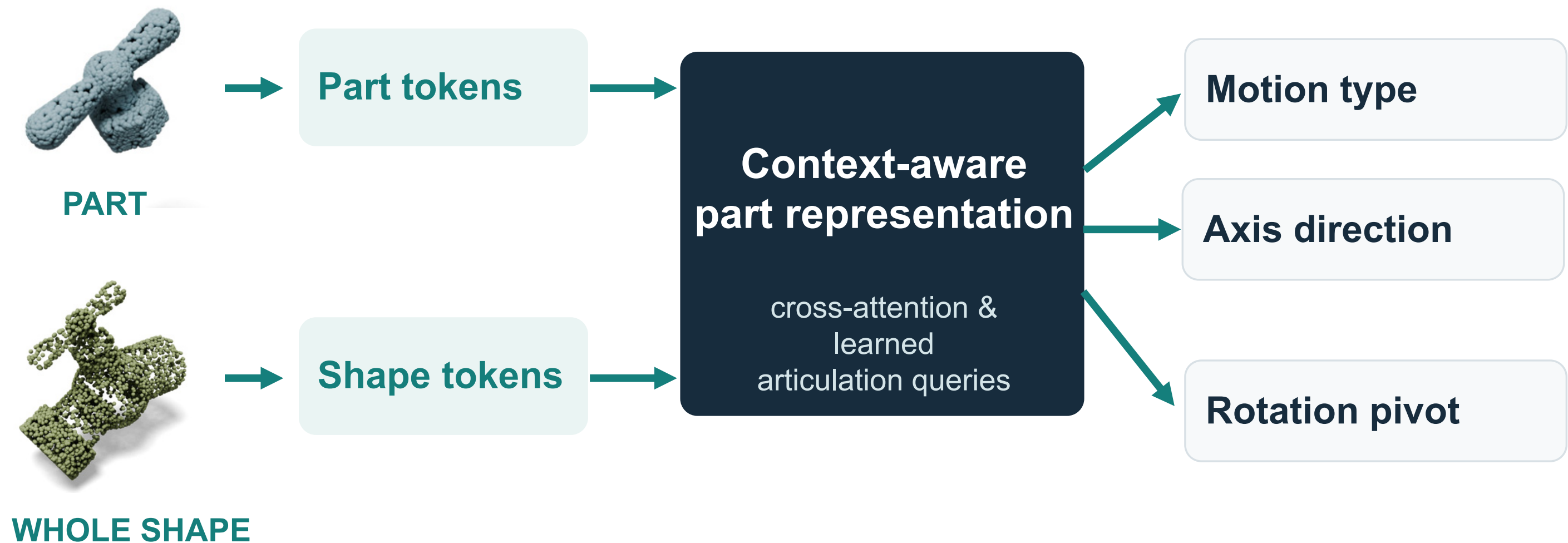


Shape tokens

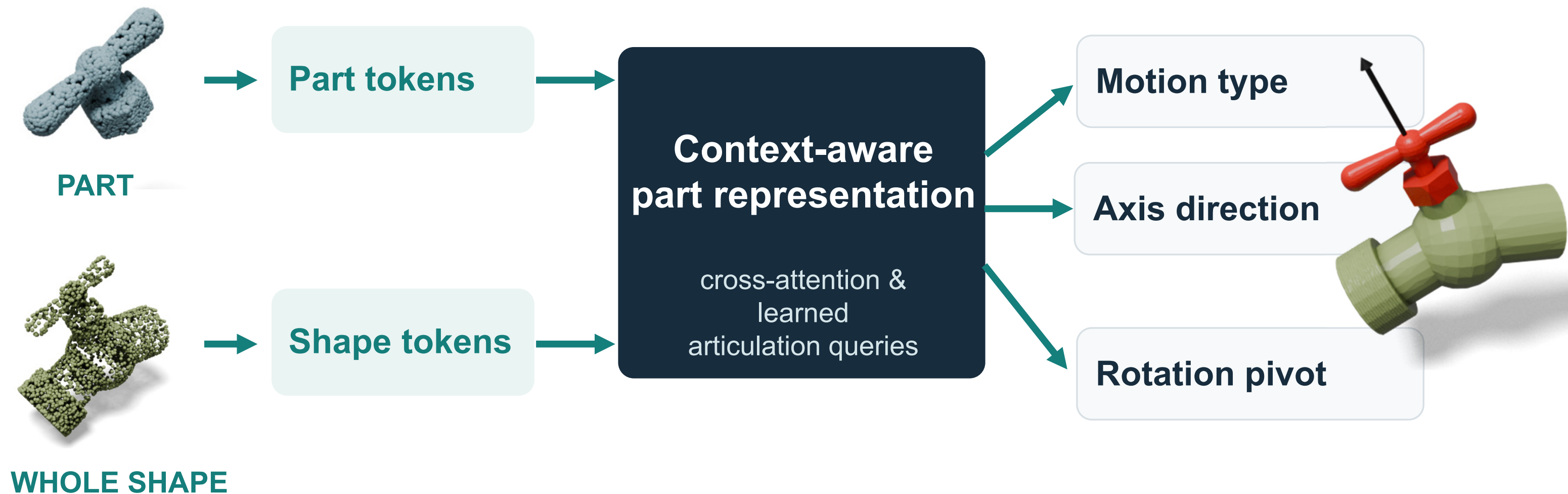
# A part makes sense in the context of its object



# A part makes sense in the context of its object



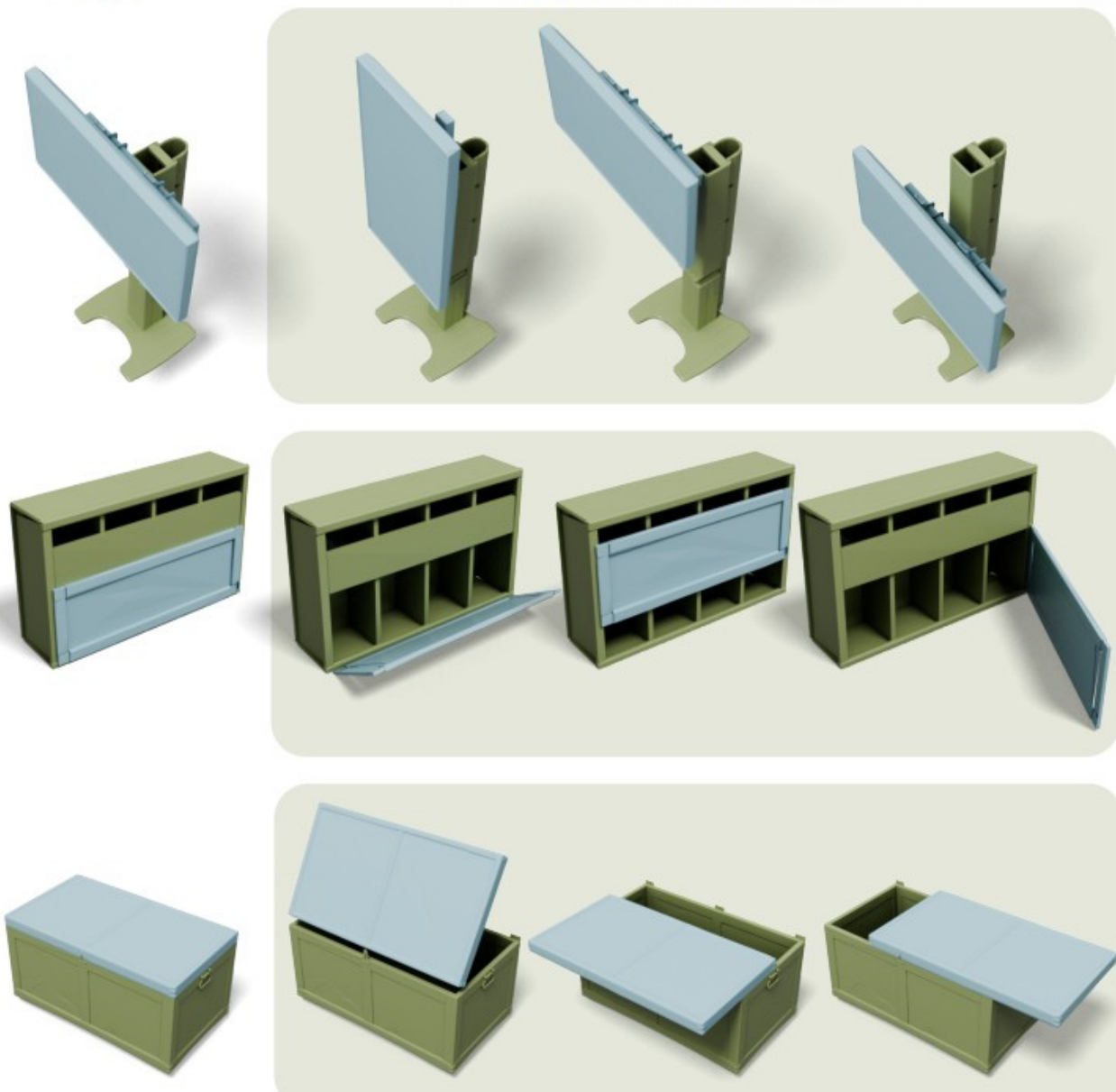
# A part makes sense in the context of its object



# Turn geometric plausibility into self-supervision

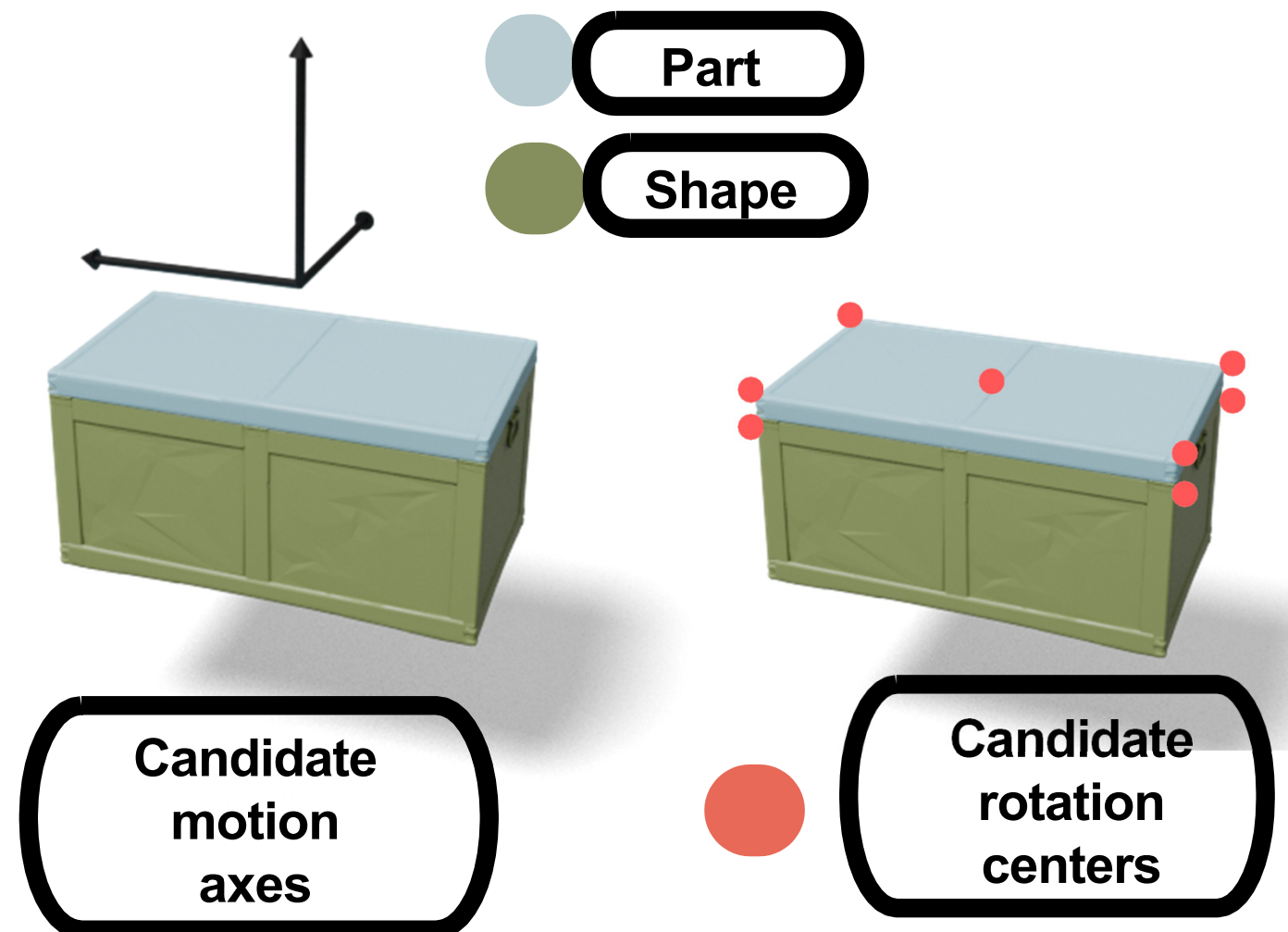
**Input**

**Candidate Articulations**



**PROPOSE**

**Principal/bbox axes + candidate pivots**

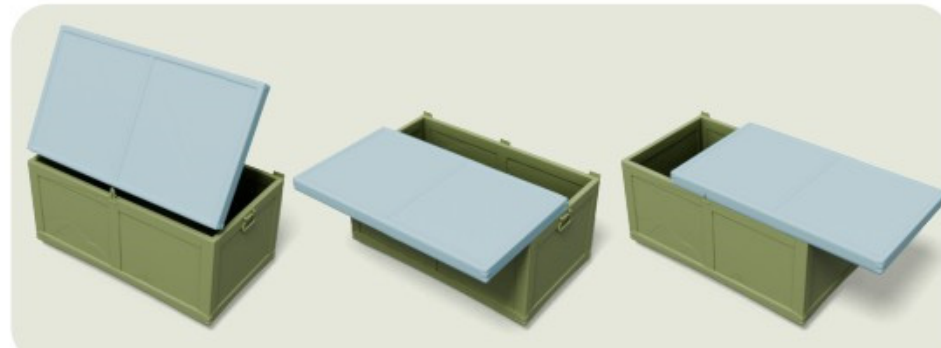
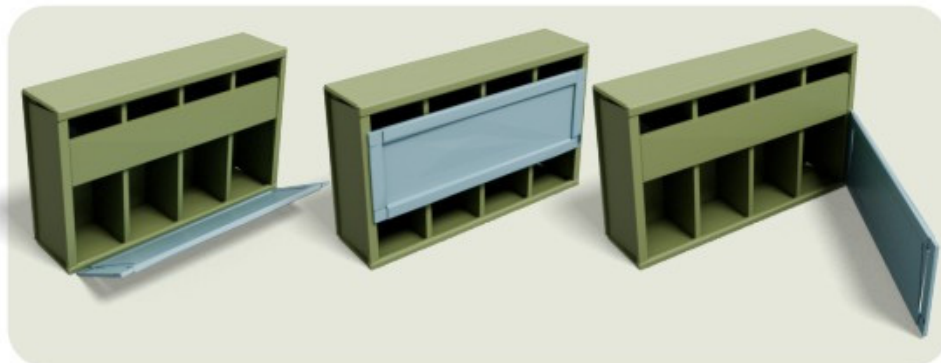
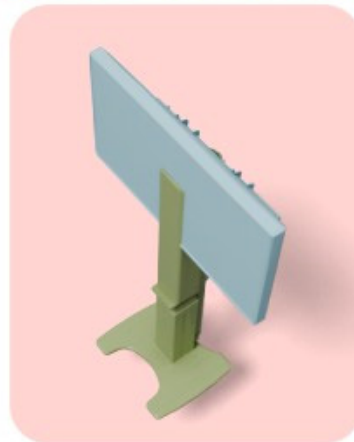
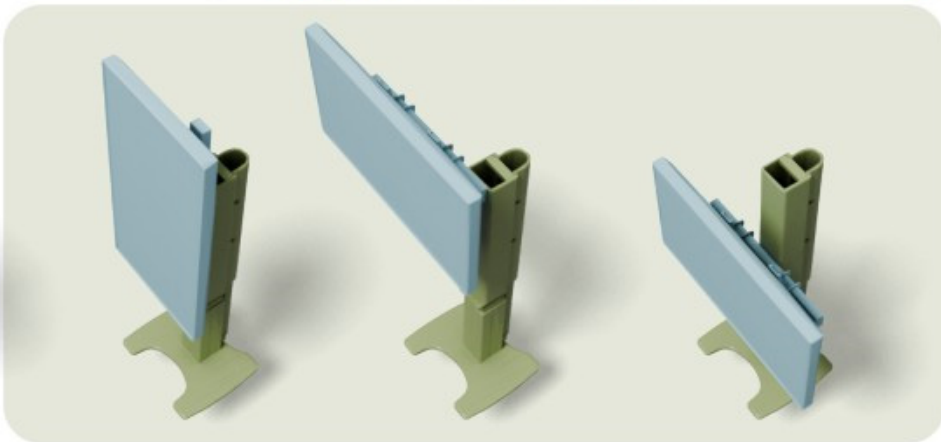
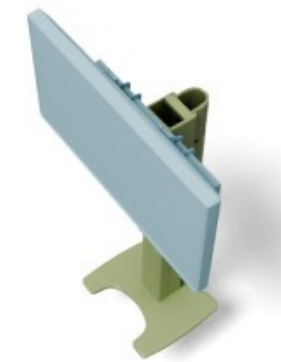


# Turn geometric plausibility into self-supervision

Input

Candidate Articulations

Invalid Articulations



PROPOSE

Principal/bbox axes + candidate pivots



TEST

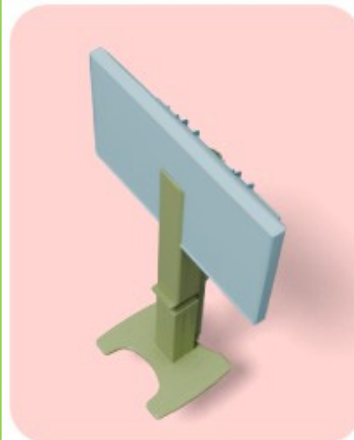
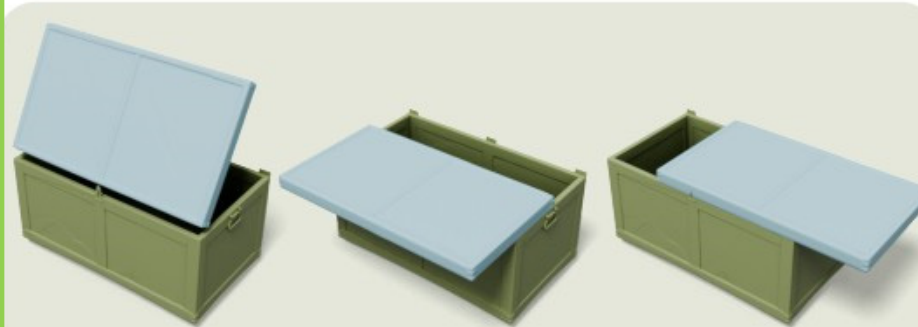
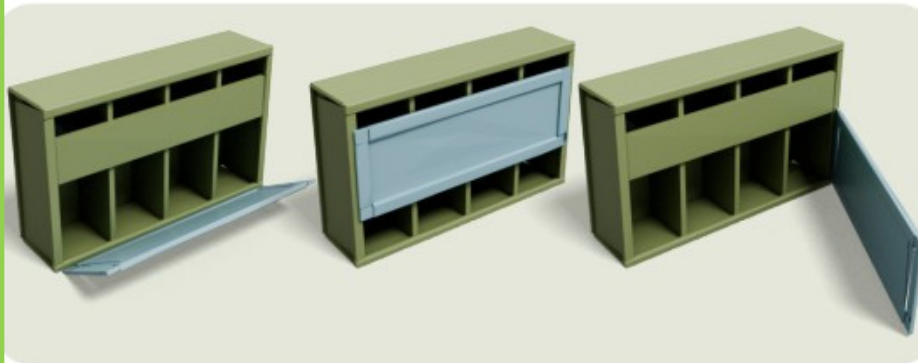
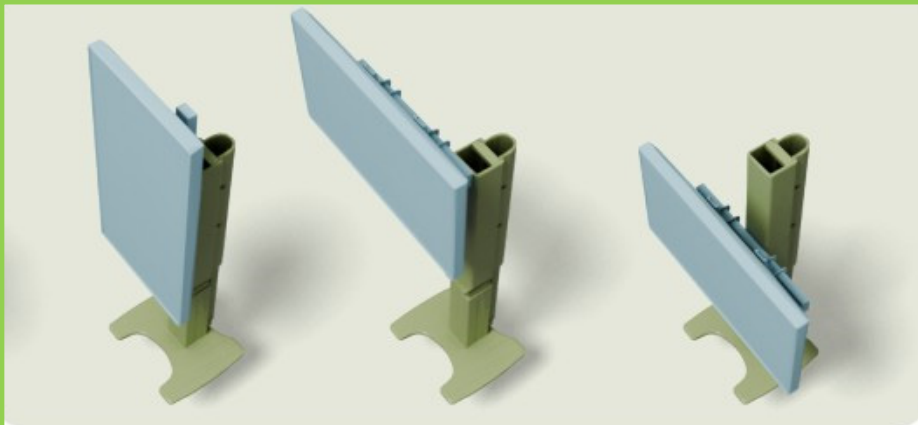
Non-trivial range;  
**reject collision / detachment**

# Turn geometric plausibility into self-supervision

Input

Candidate Articulations

Invalid Articulations



PROPOSE

Principal/bbox axes + candidate pivots



TEST

Non-trivial range;  
**reject collision / detachment**



LEARN

**Pretrain on candidate articulations,  
then fine-tune on labeled ones**

# Pre-training improves articulation prediction

**Axis error ↓**

PartNet-Mobility | no semantic part labels

No pretraining A horizontal bar chart where the bar is divided into two segments: a longer dark gray segment on the left and a shorter light gray segment on the right, representing a higher axis error.

**w/ pretraining** A horizontal bar chart where the bar is divided into two segments: a longer dark teal segment on the left and a shorter light gray segment on the right, representing a lower axis error compared to the 'No pretraining' case.

**~14% lower**

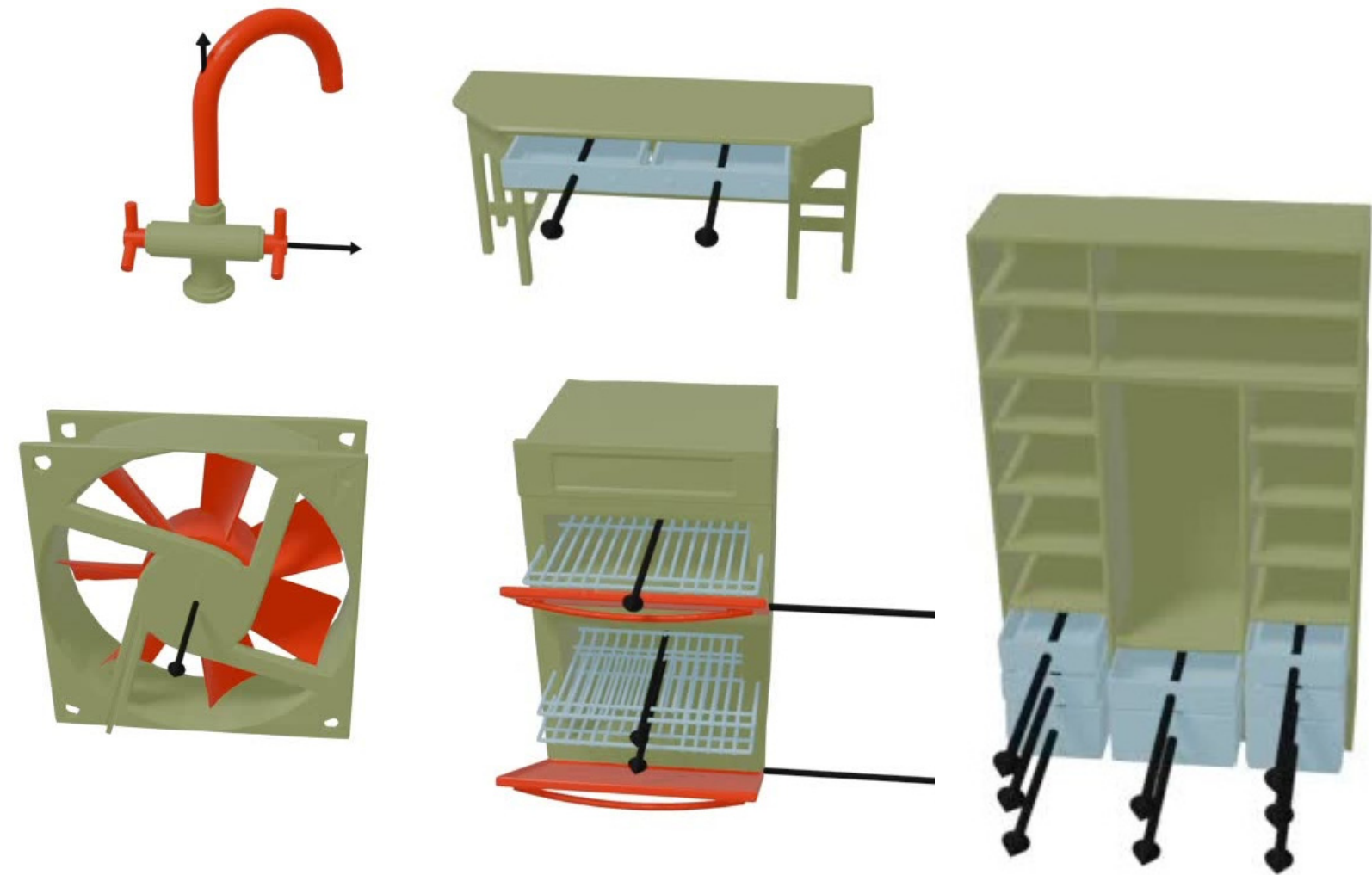
# Pre-training improves articulation prediction

**Axis error ↓**

PartNet-Mobility | no semantic part labels

No pretraining 

w/ pretraining   
~14% lower



PREDICTED ARTICULATIONS

# Roadmap

1

## GEOPARD

What articulations are plausible for given assets?

2

## Rigel3D

How to generate assets with their articulation?

3

## SOPHY

How should assets respond in a physical world?

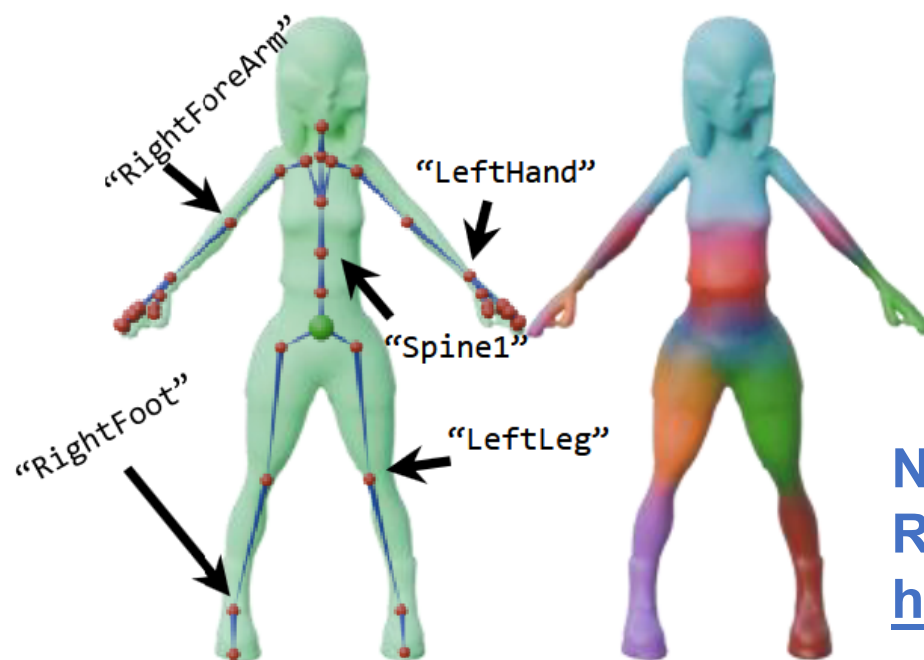
4

## BLARM

How can we drive the motion of assets?

Skeleton & Labels

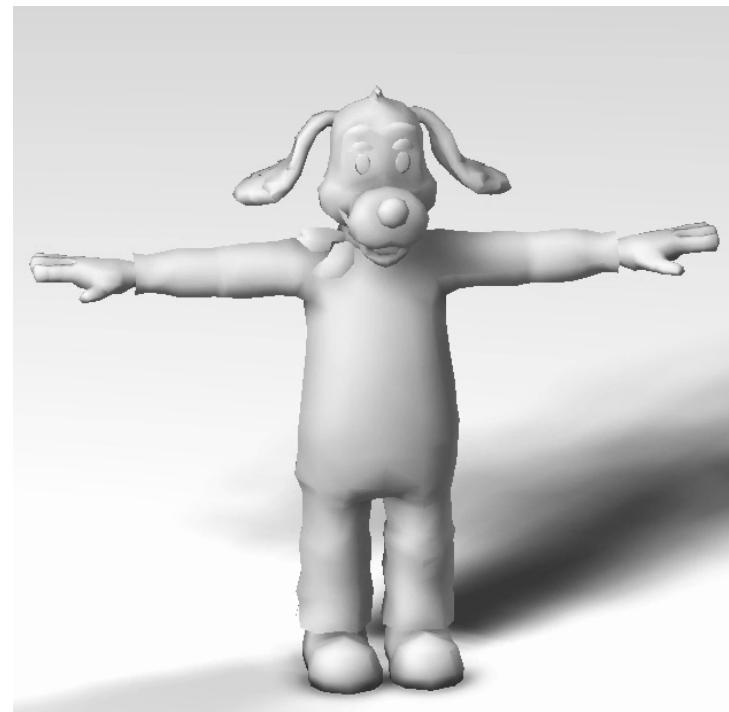
Skin Weights



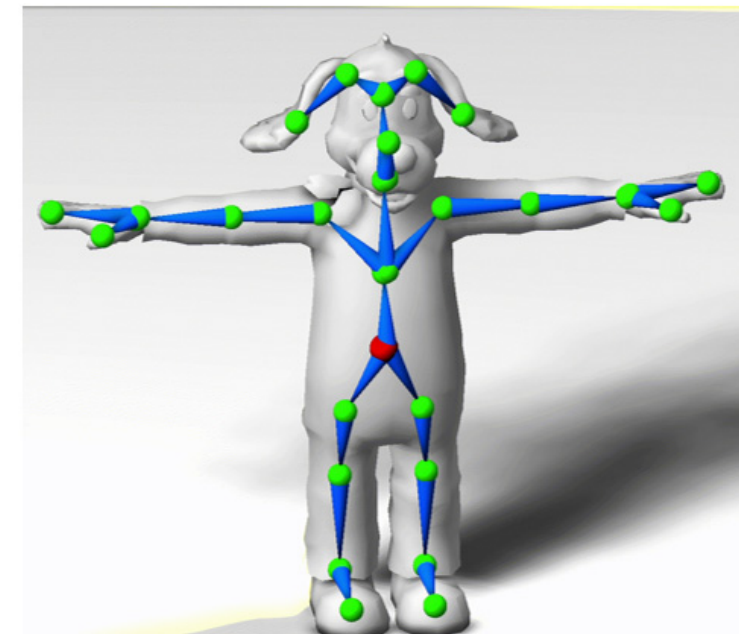
Nikitas Chatzis, M. Loizou, E. Kalogerakis  
Rigel3D: Rig-aware Latents for Animation-Ready 3D Asset Generation  
<https://arxiv.org/abs/2605.13129>

# Attach a rig after generating the shape?

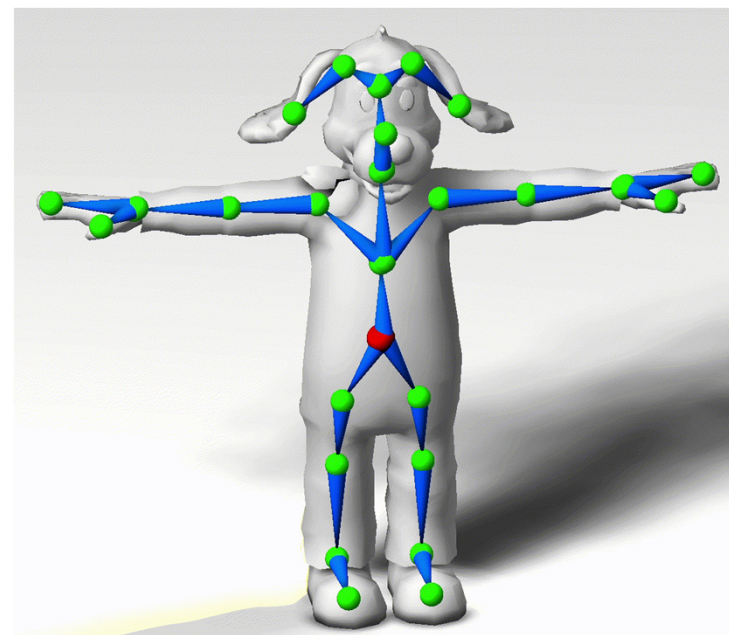
## POST-HOC PIPELINE



Input 3D asset



Skeleton



Skinning

Strong post-hoc auto-rigging literature

[RigNet, 2020](#)

[MoRig, 2022](#)

[TARig, 2023](#)

[Anymate, 2025](#)

[MagicArticulate, 2025](#)

[Puppeteer, 2025](#)

...

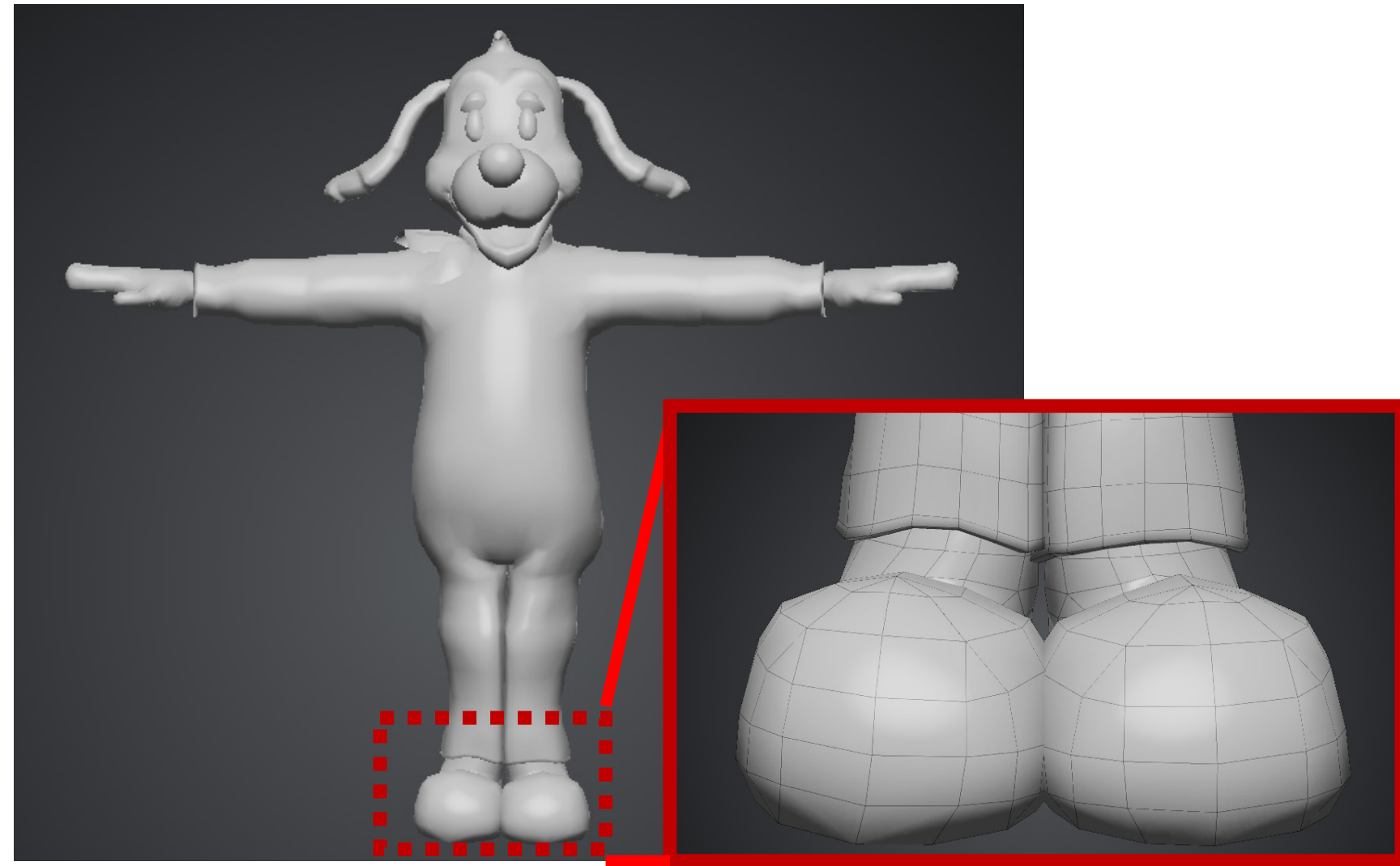
# Attach a rig after generating the shape?

## POST-HOC PIPELINE



### Potential issues:

- Generation errors become constraints for the downstream rigger
- Decouples geometry from motion control



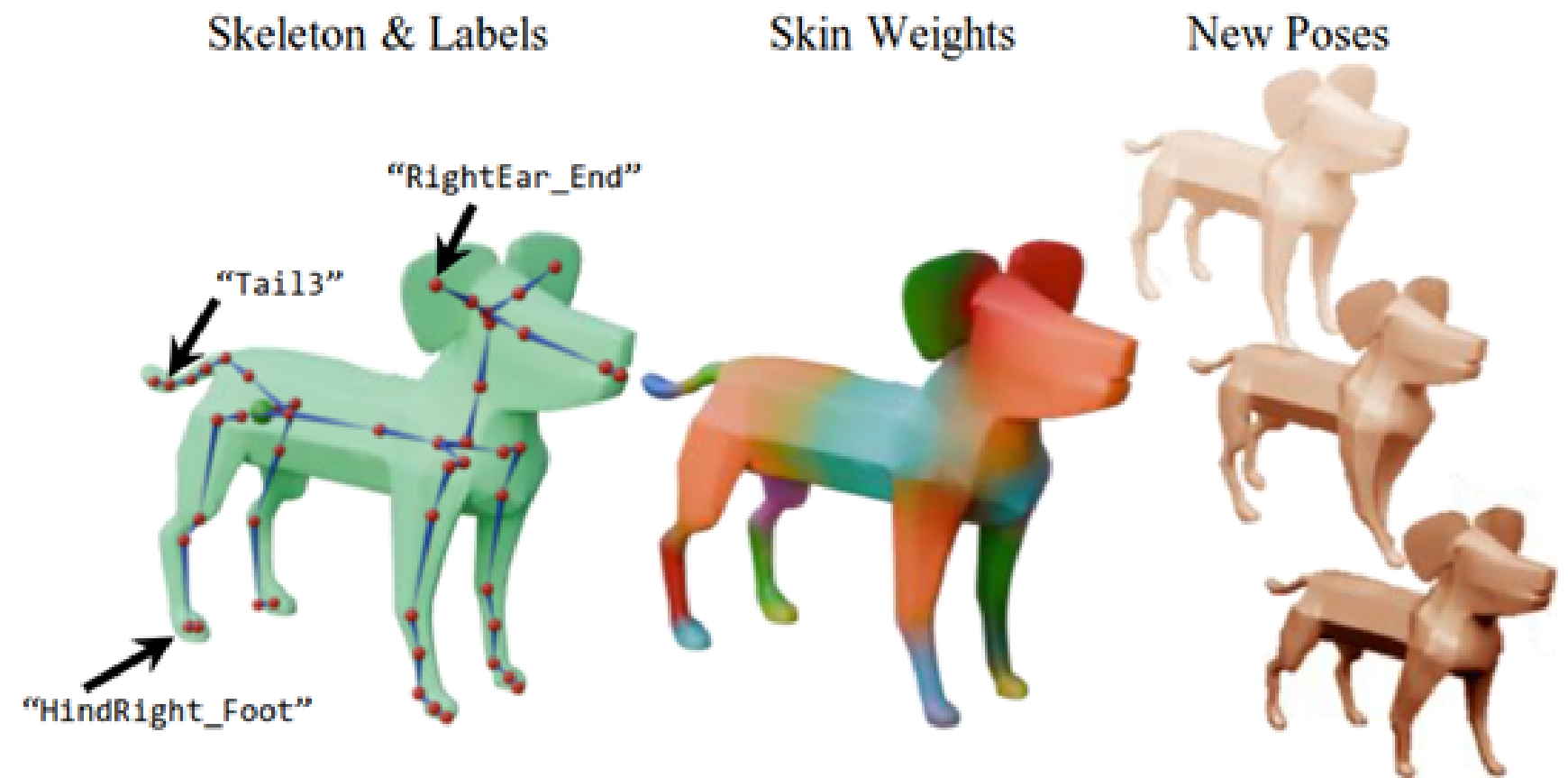
Generated 3D asset  
(Illustrative generation failure)

# Rigel3D: Generate geometry & rigs jointly

## POST-HOC PIPELINE



## JOINT MODELING PIPELINE



# Two structured latent spaces, one asset

## **SURFACE LATENTS**

Geometry + appearance near the boundary

## **SKELETON LATENTS**

Sparse internal articulation structure

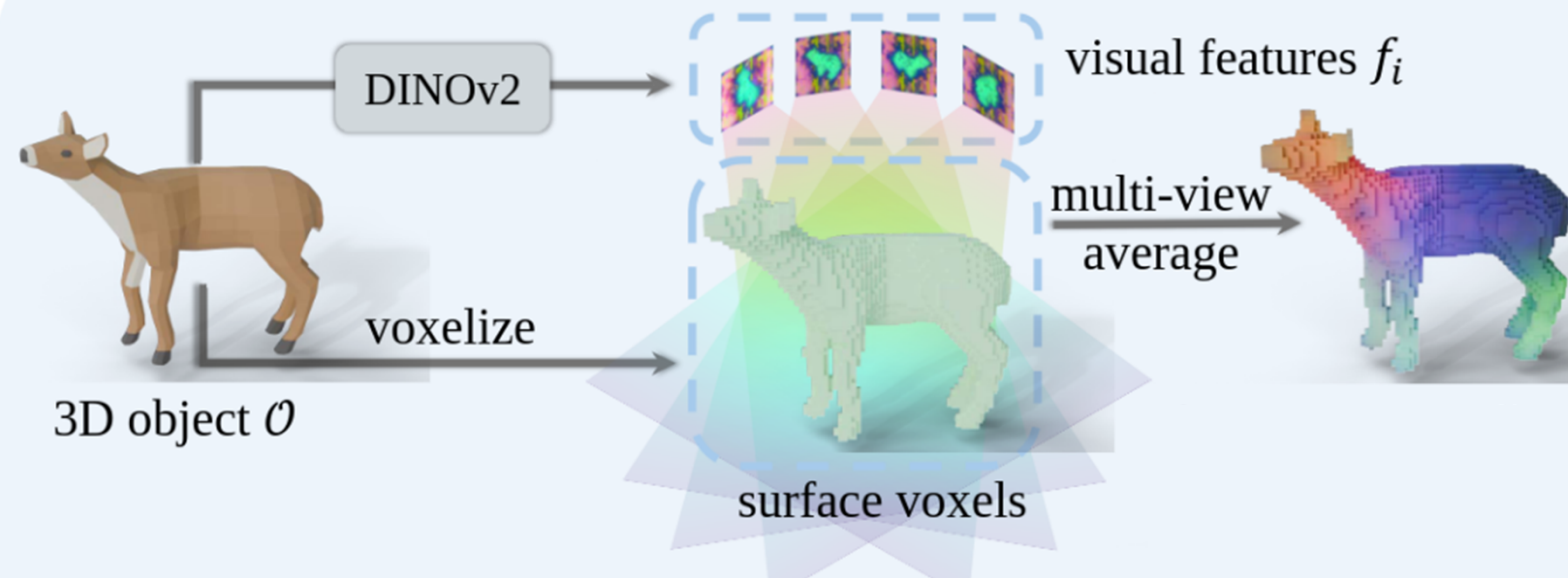
# Two structured latent spaces, one asset

## SURFACE LATENTS

Geometry + appearance near the boundary

## SKELETON LATENTS

Sparse internal articulation structure



[TRELLIS, CVPR 2025]

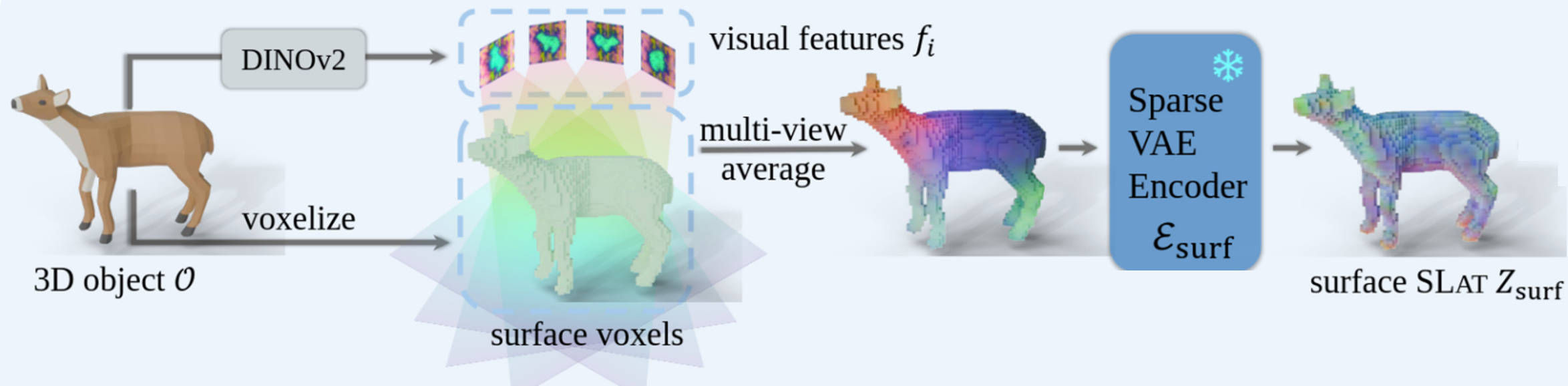
# Two structured latent spaces, one asset

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## SKELETON LATENTS

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[TRELLIS, CVPR 2025]

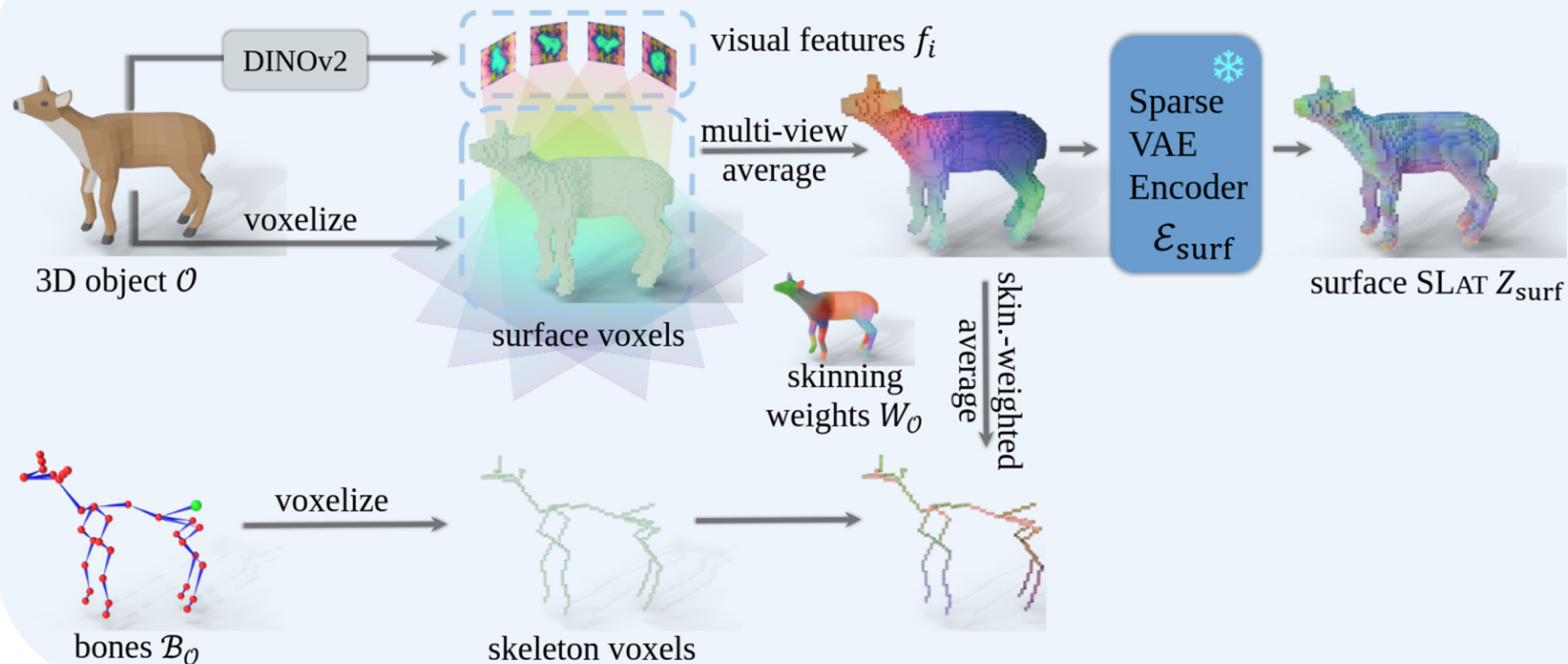
# Two structured latent spaces, one asset

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Geometry + appearance near the boundary

## SKELETON LATENTS

Sparse internal articulation structure



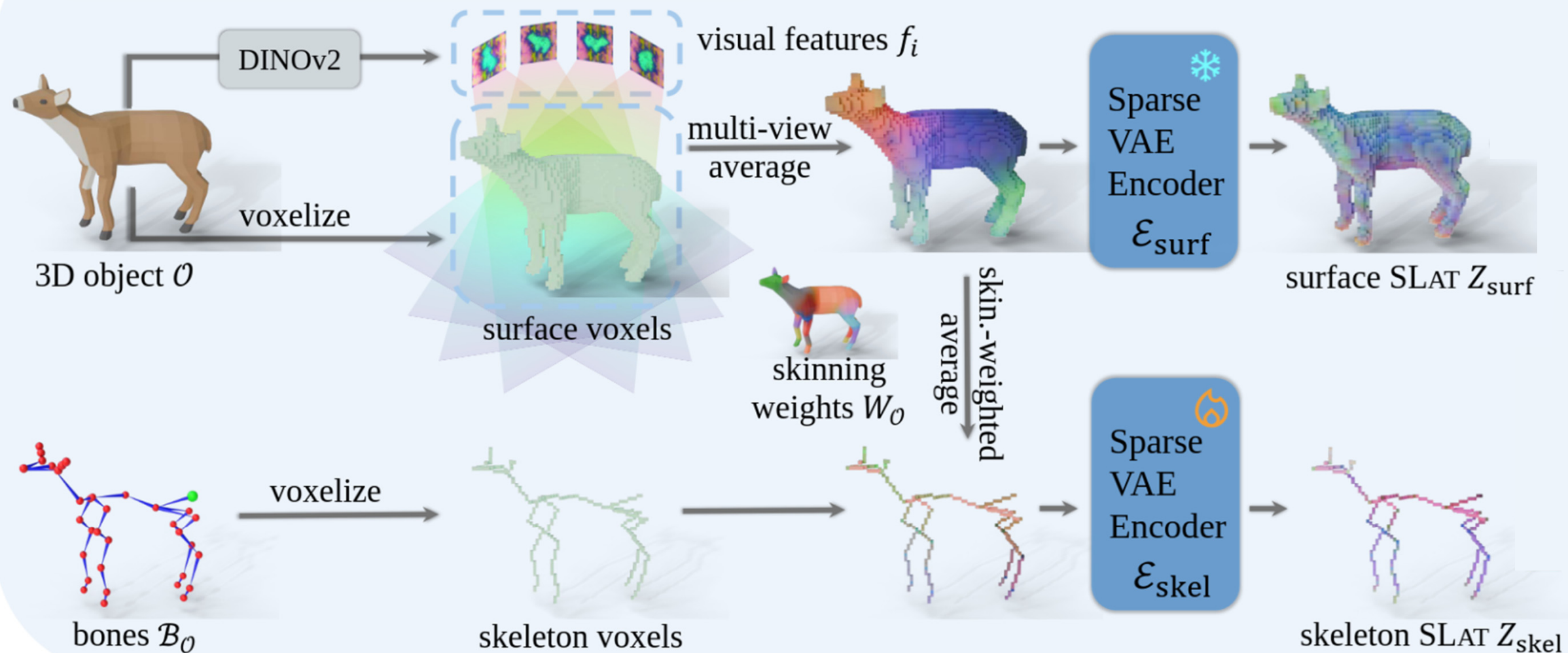
# Two structured latent spaces, one asset

## SURFACE LATENTS

Geometry + appearance near the boundary

## SKELETON LATENTS

Sparse internal articulation structure



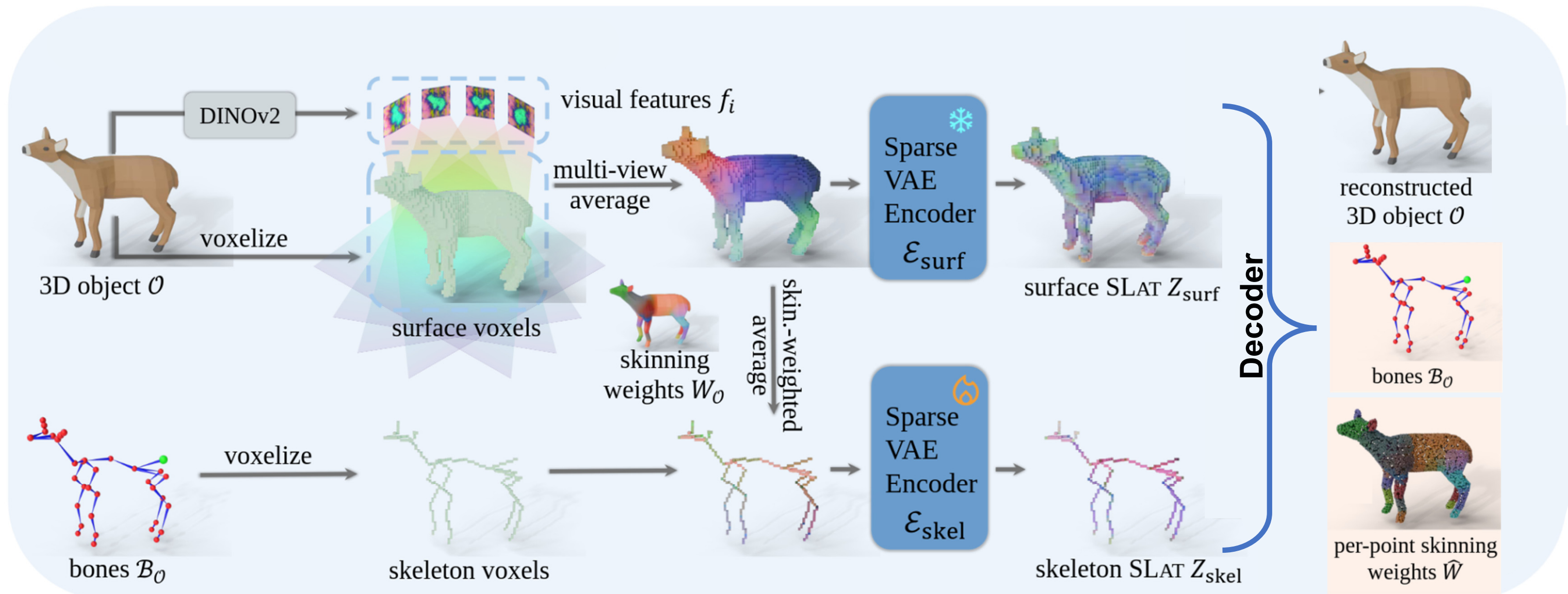
# Two structured latent spaces, one asset

## SURFACE LATENTS

Geometry + appearance near the boundary

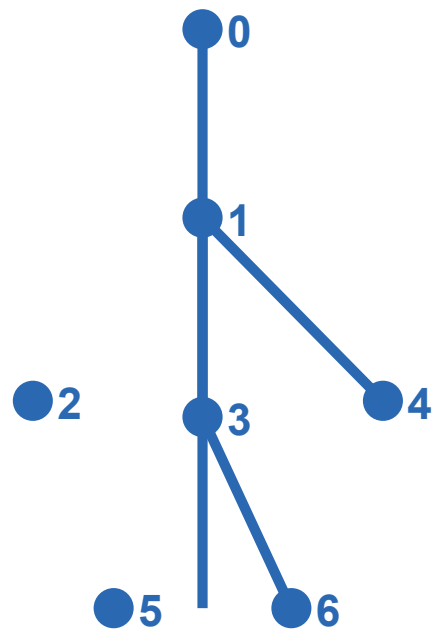
## SKELETON LATENTS

Sparse internal articulation structure



# Generate joints autoregressively, then bind the surface

## Variable-size hierarchy

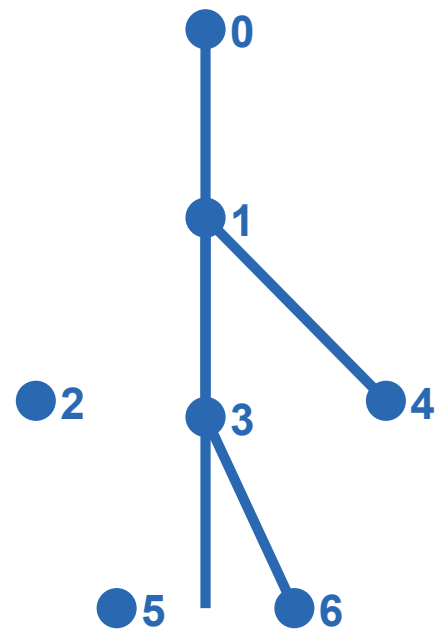


## Autoregressive joints

$$t_k = (q_x^k, q_y^k, q_z^k, \pi_k)$$

# Generate joints autoregressively, then bind the surface

## Variable-size hierarchy



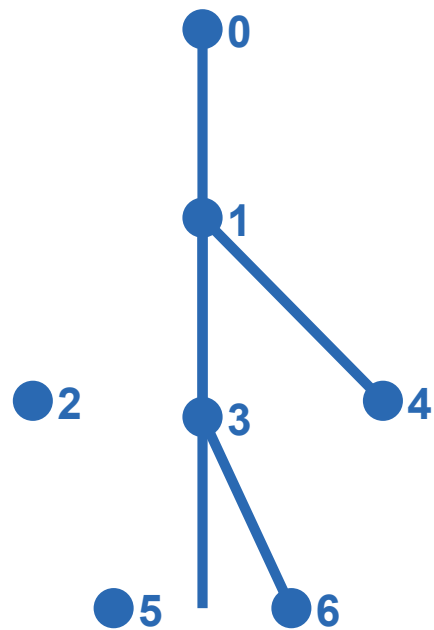
## Autoregressive joints

$$t_k = (q_x^k, q_y^k, q_z^k, \pi_k)$$

3D position

# Generate joints autoregressively, then bind the surface

## Variable-size hierarchy



## Autoregressive joints

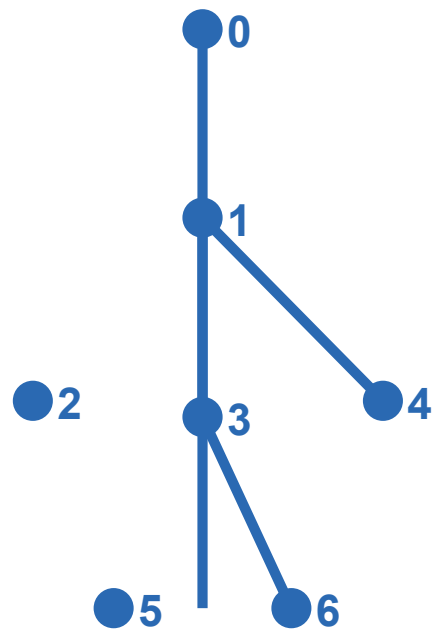
$$t_k = (q_x^k, q_y^k, q_z^k, \pi_k)$$

parent  
index

parent index < current index

# Generate joints autoregressively, then bind the surface

## Variable-size hierarchy



## Autoregressive joints

$$t_k = (q_x^k, q_y^k, q_z^k, \pi_k)$$

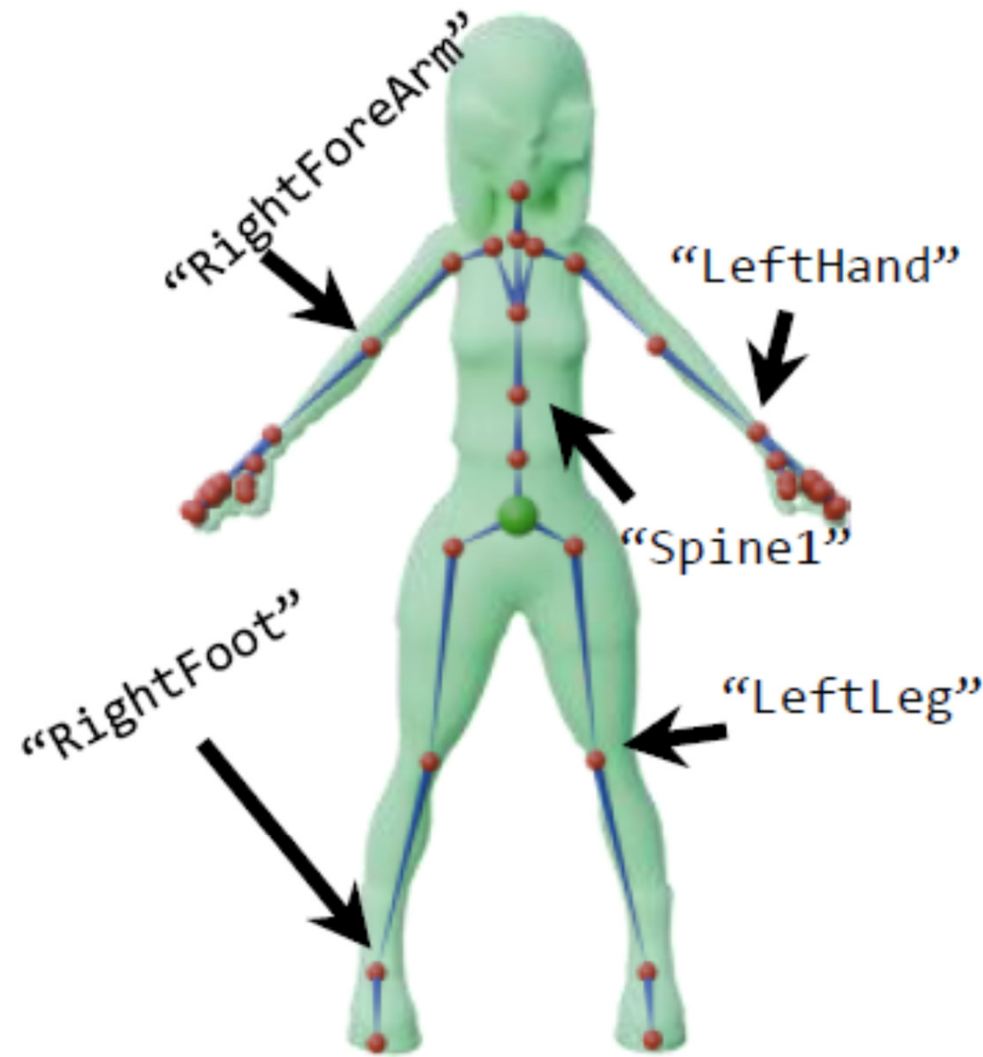
parent index < current index

## Bind surface to bones



Point–bone attention  
→ skinning weights

# Joint labeling => transfer motion from diverse templates

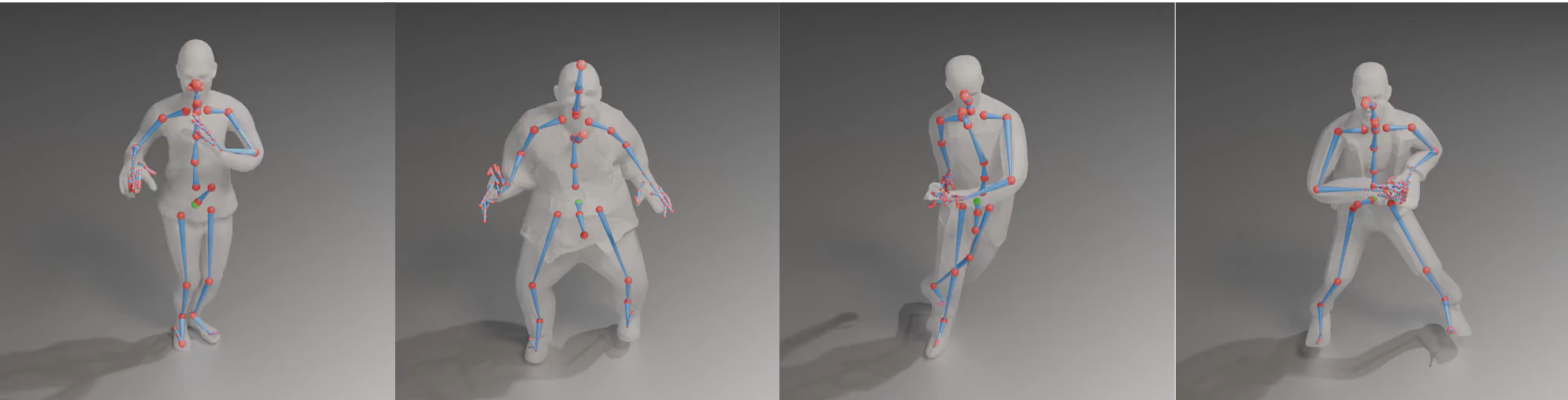


**Joint embedding**  
Joint features  $\leftrightarrow$  text features

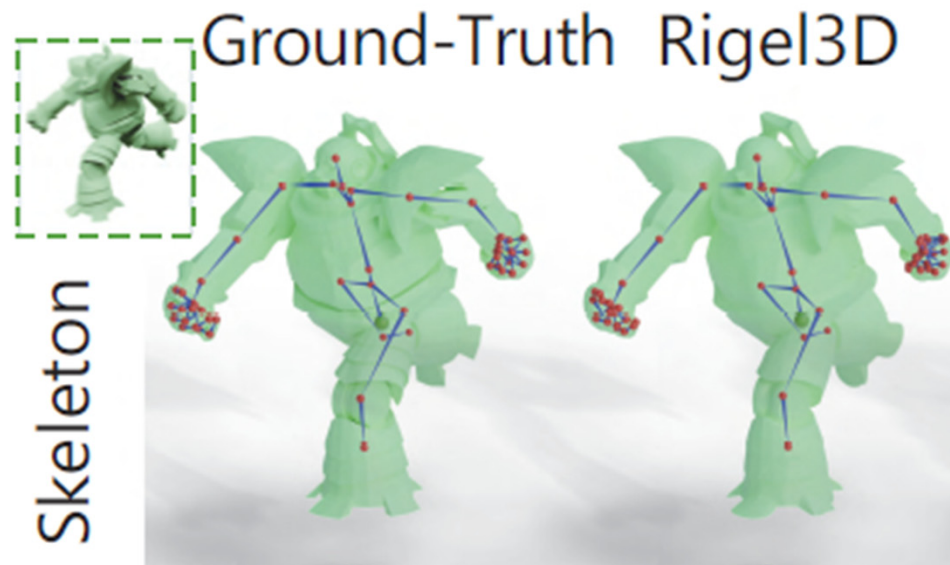


**Open-vocabulary joint labeling**  
Match to candidate joint names

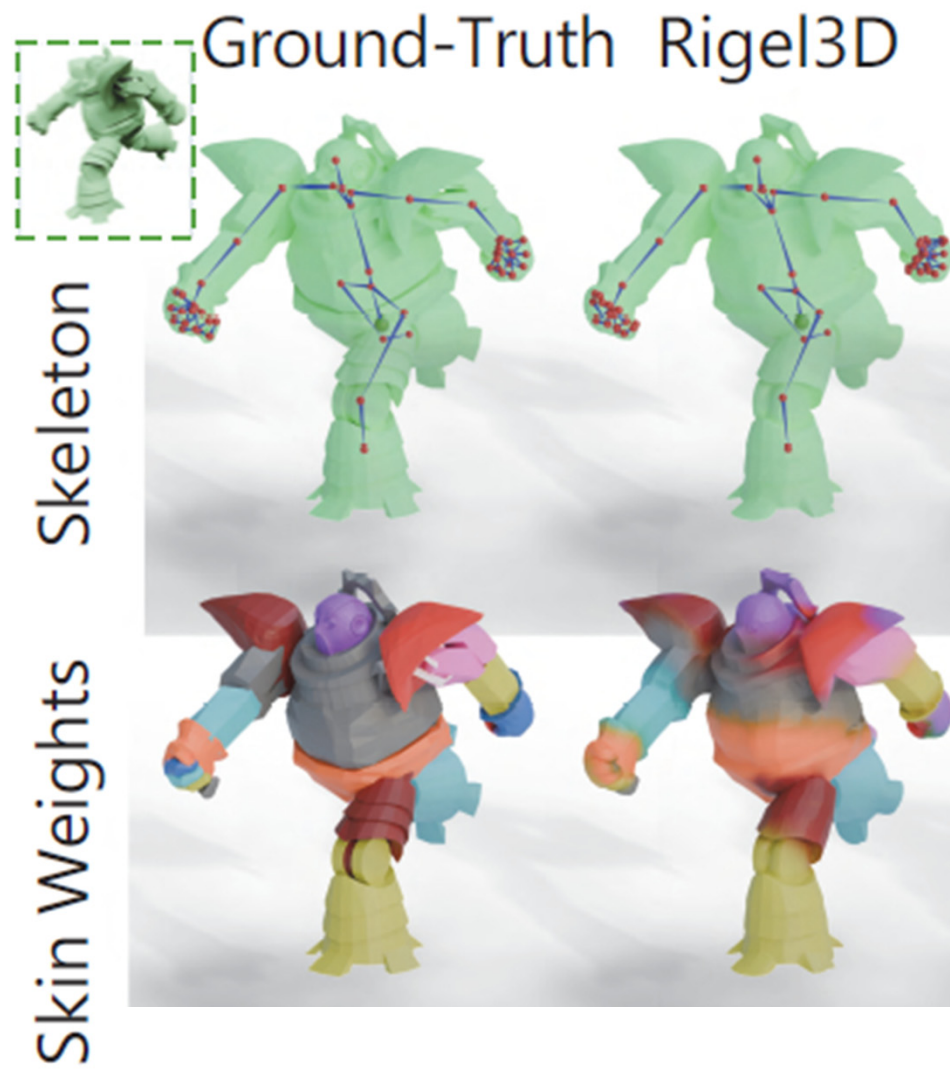
# Automatic motion retargeting from templates



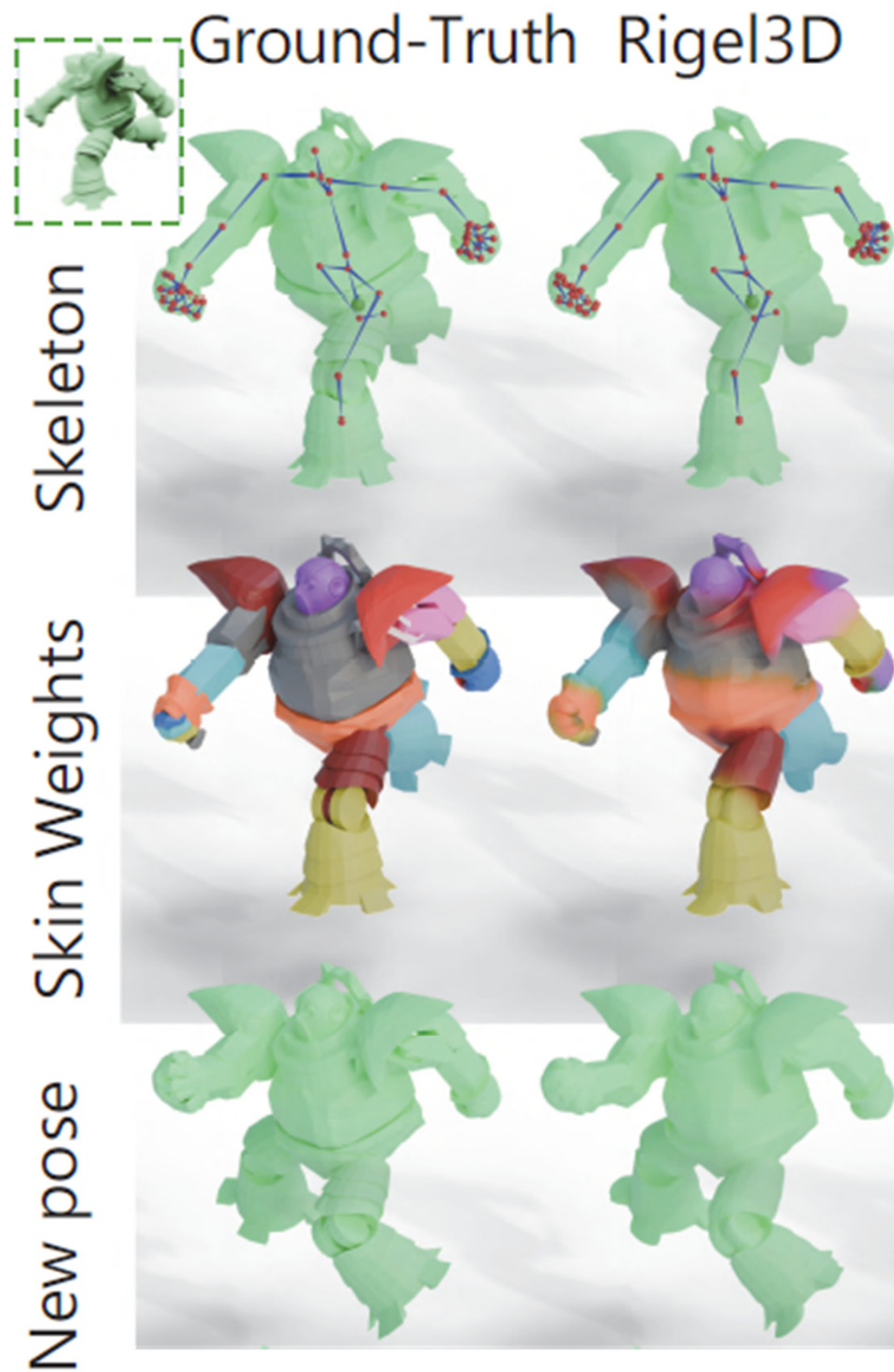
# Joint Shape-Rig generation helps



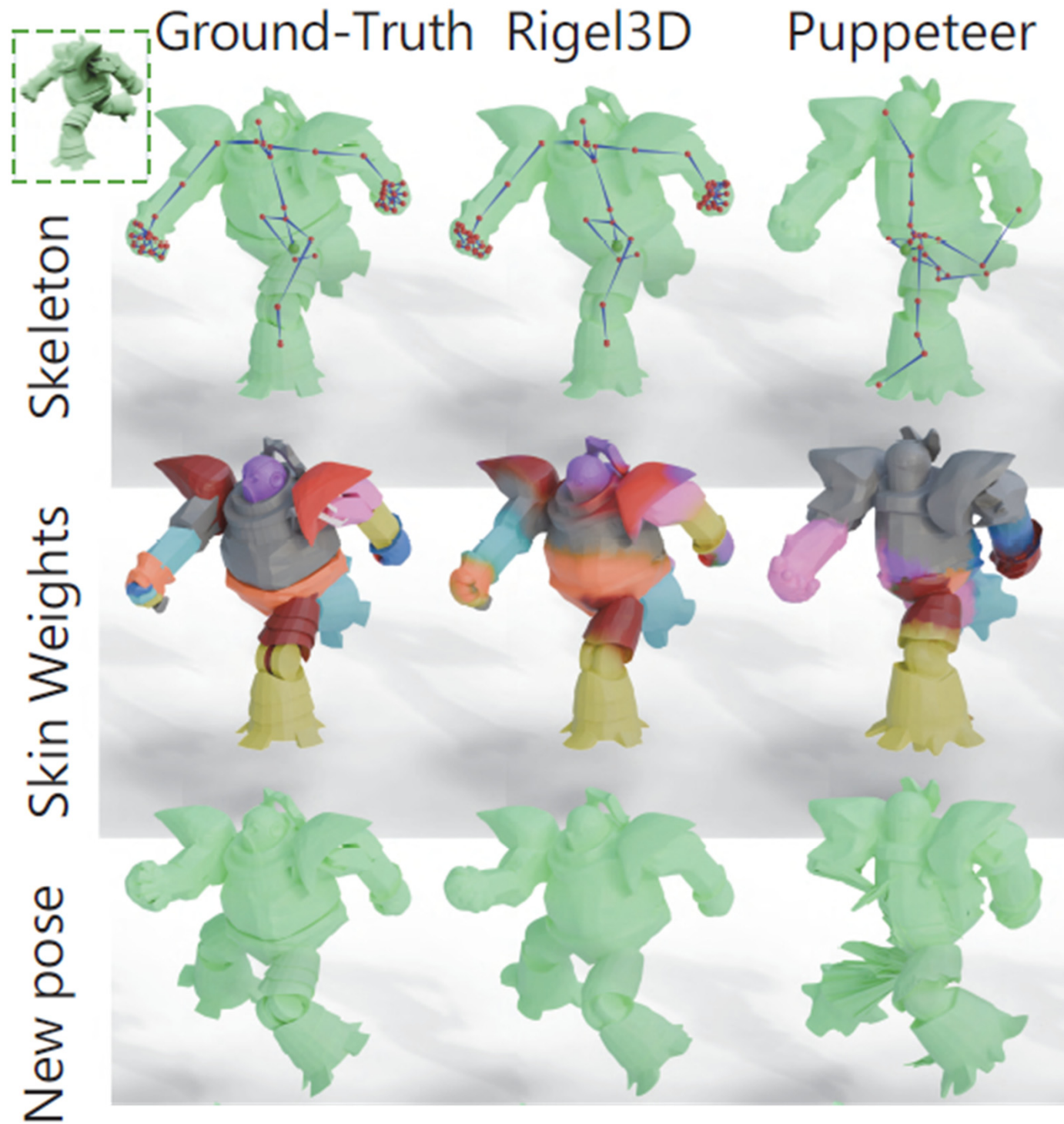
# Joint Shape-Rig generation helps



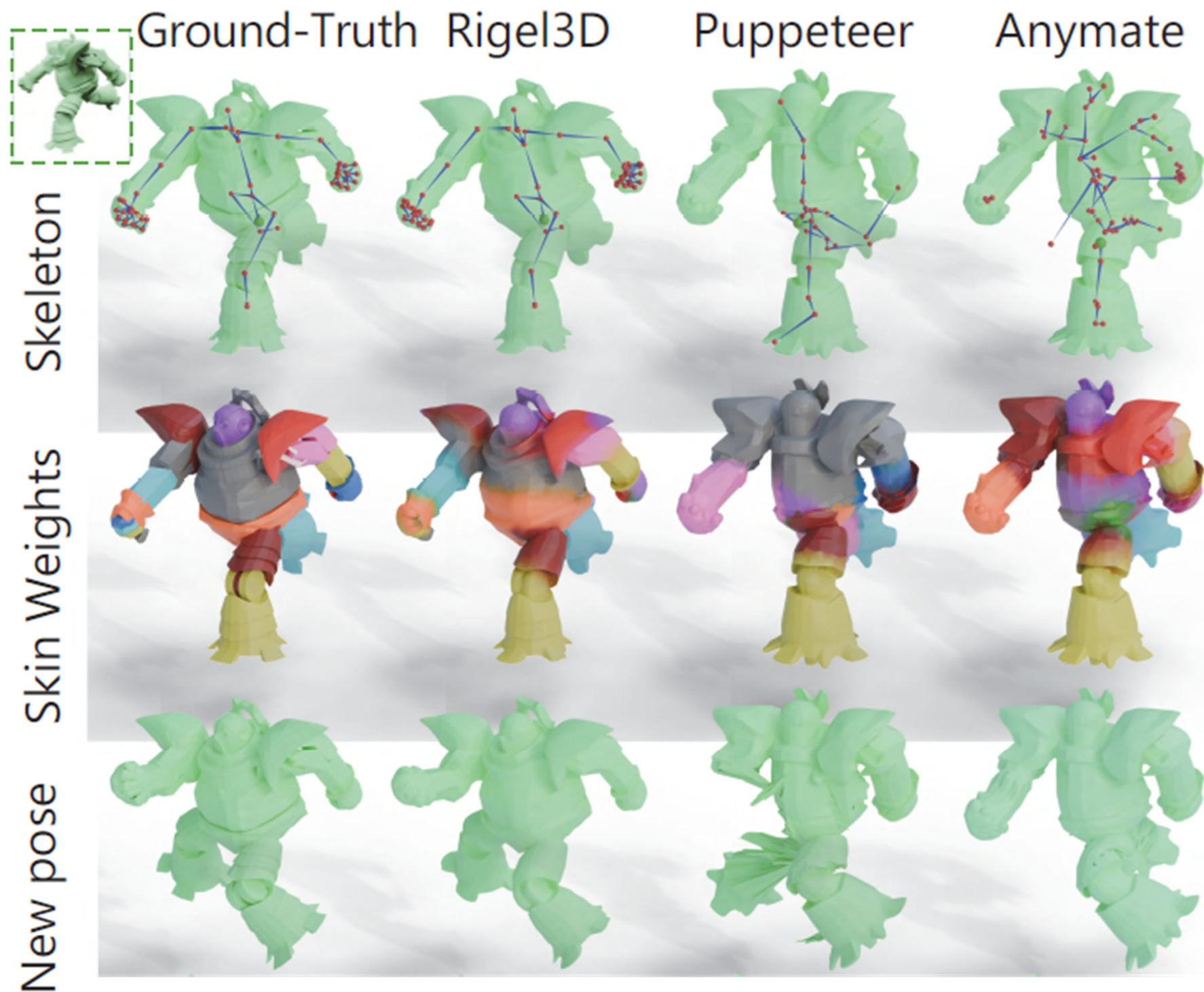
# Joint Shape-Rig generation helps



# Joint Shape-Rig generation helps



# Joint Shape-Rig generation helps



**Bone-to-bone error ↓**

Evaluated on Anymate test split

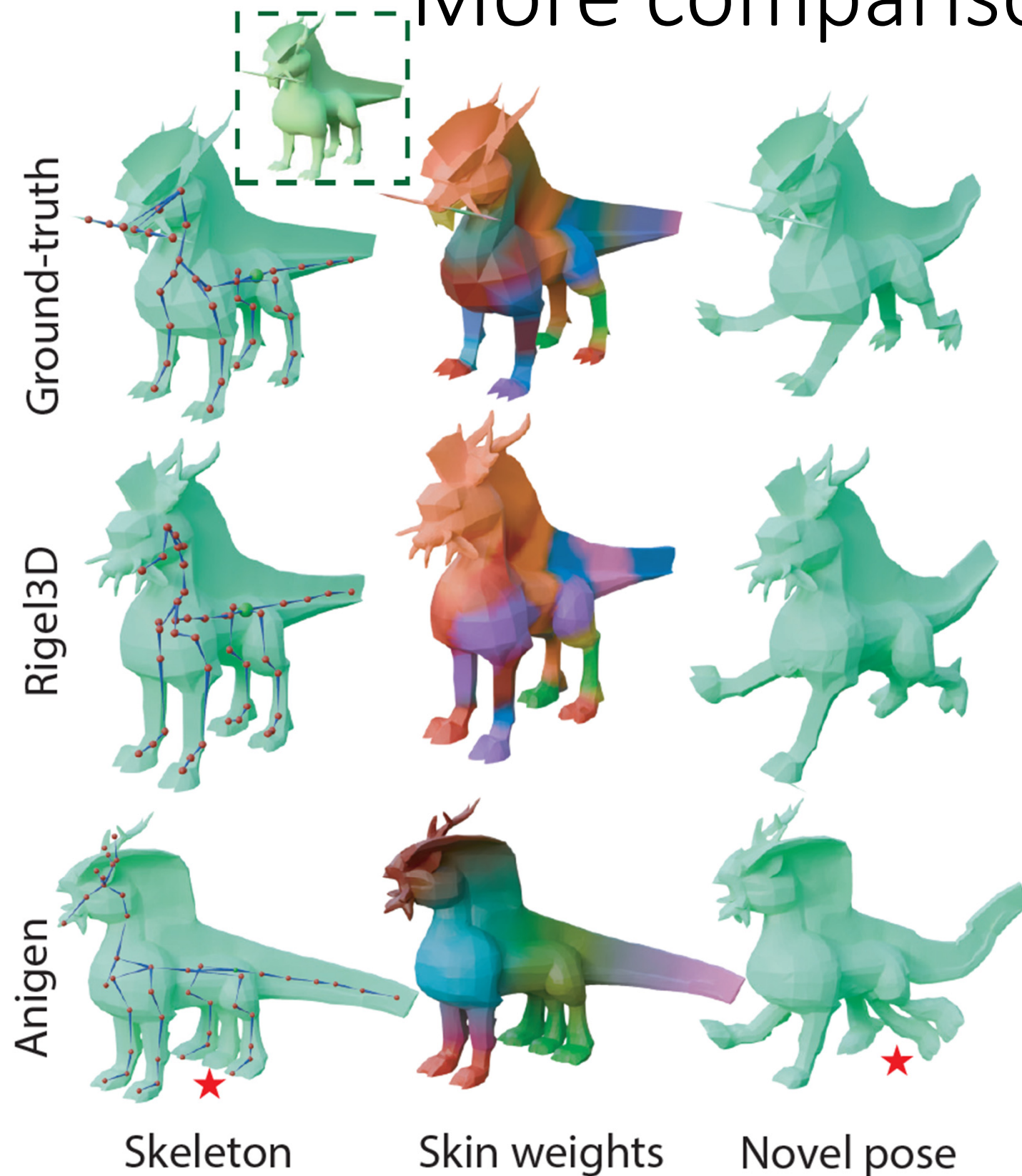
Puppeteer 5.7

Anymate 5.4

**Rigel3D 4.9**

**~10% lower**

# More comparisons with SOTA



**Bone-to-bone error ↓**

Evaluated on ArticulationXL test split

Anigen 8.6

Rigel3D 6.0

**~30% lower**

# Roadmap

1

## GEOPARD

What articulations are plausible for given assets?

2

## Rigel3D

How to generate assets with their articulation?

3

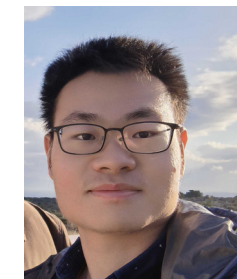
## SOPHY

How should assets respond in a physical world?

4

## BLARM

How can we drive the motion of assets?



Junyi Cao, E. Kalogerakis, WACV 2026

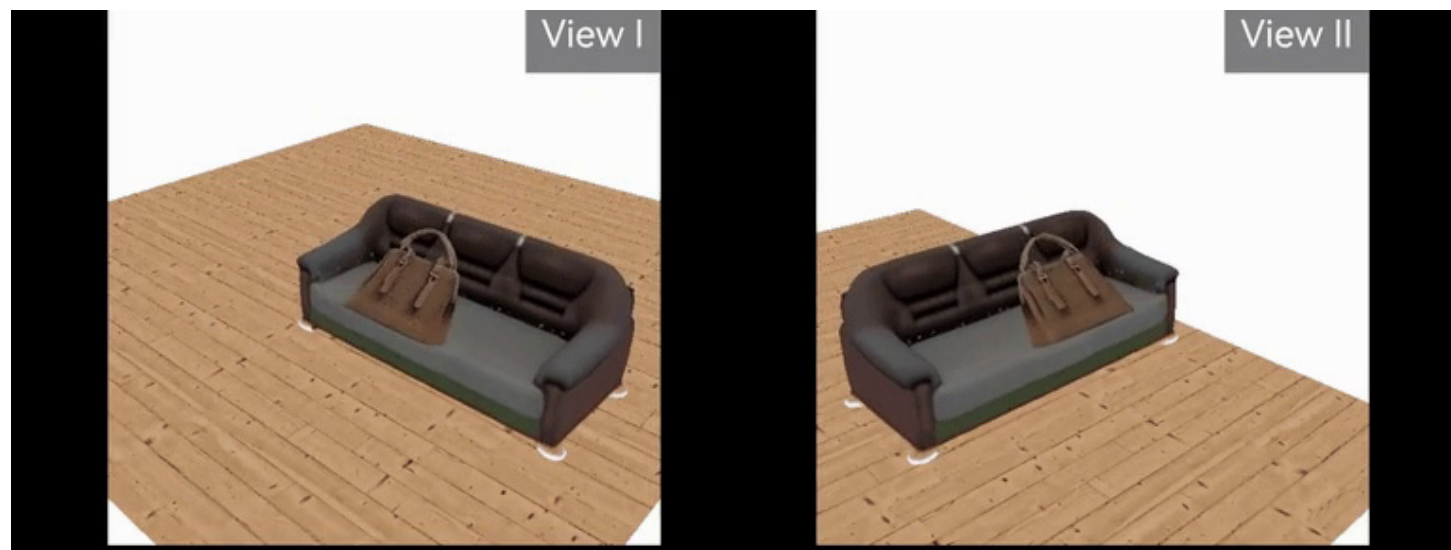
SOPHY: Generating Simulation-Ready  
Objects with PHYSical Materials

[https://xjay18.github.io/SOPHY\\_page/](https://xjay18.github.io/SOPHY_page/)

# A rig is not a physical model

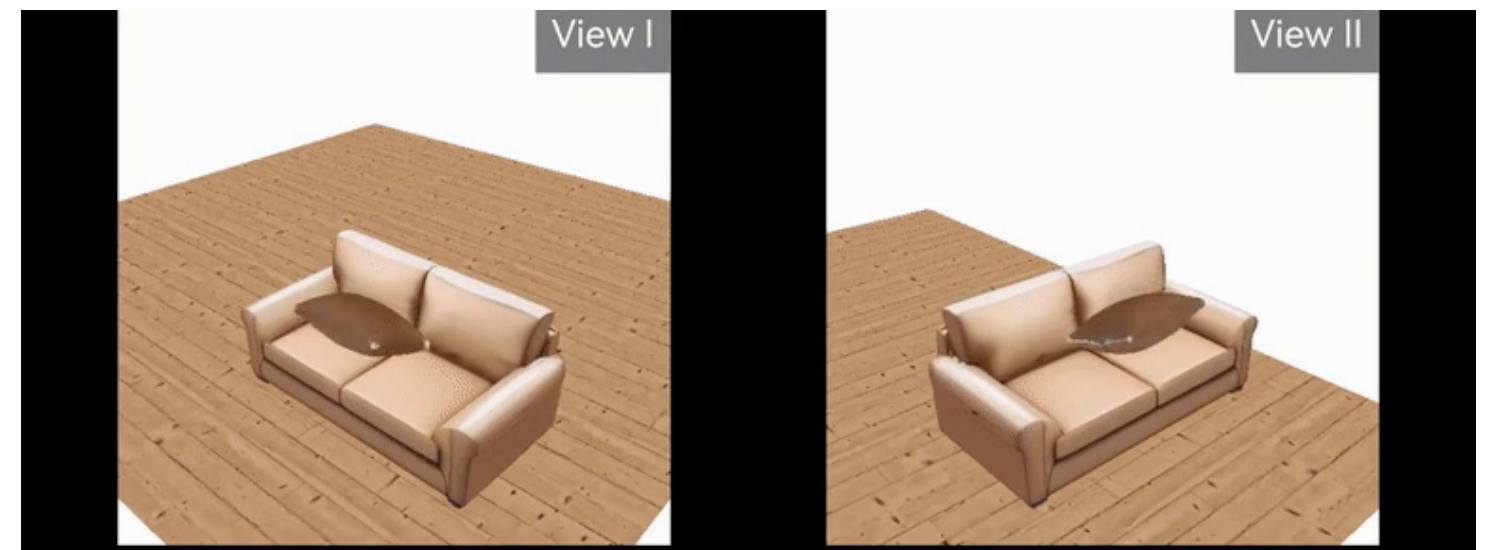
We need **physical** properties, not just articulation structures and rendering materials.

Generated bag



PHYSICS SIMULATION

Generated loveseat



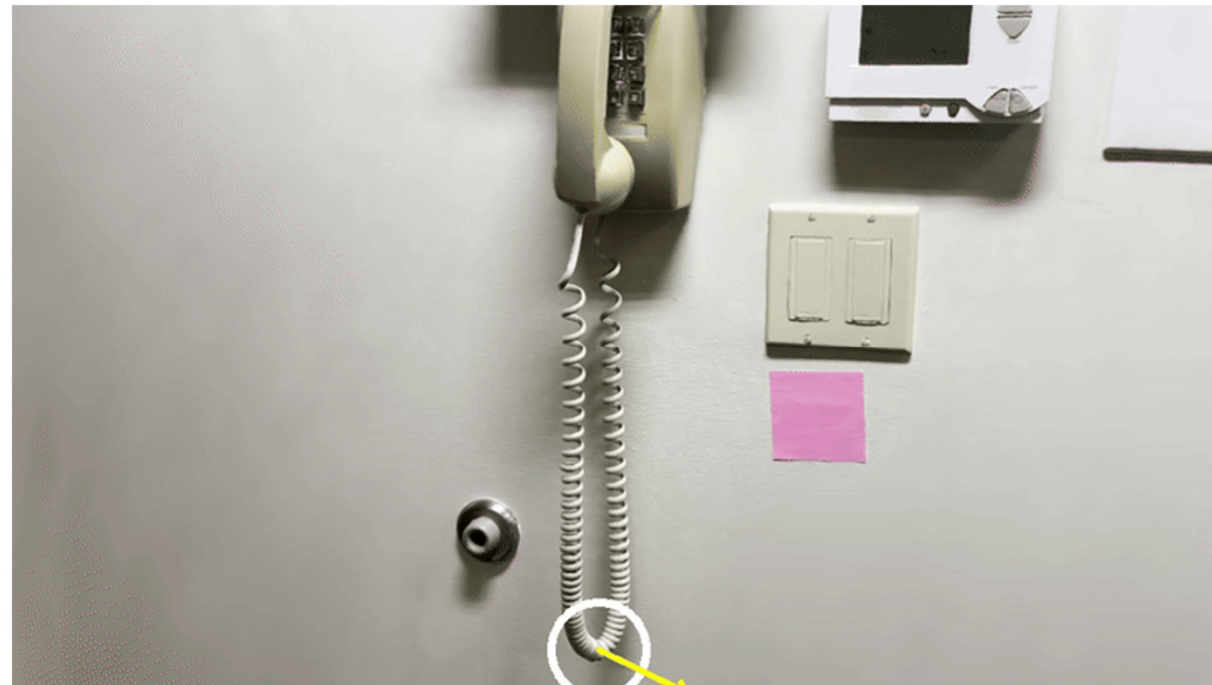
PHYSICS SIMULATION

Generated geometry + material fields → simulator under specified forces and conditions.

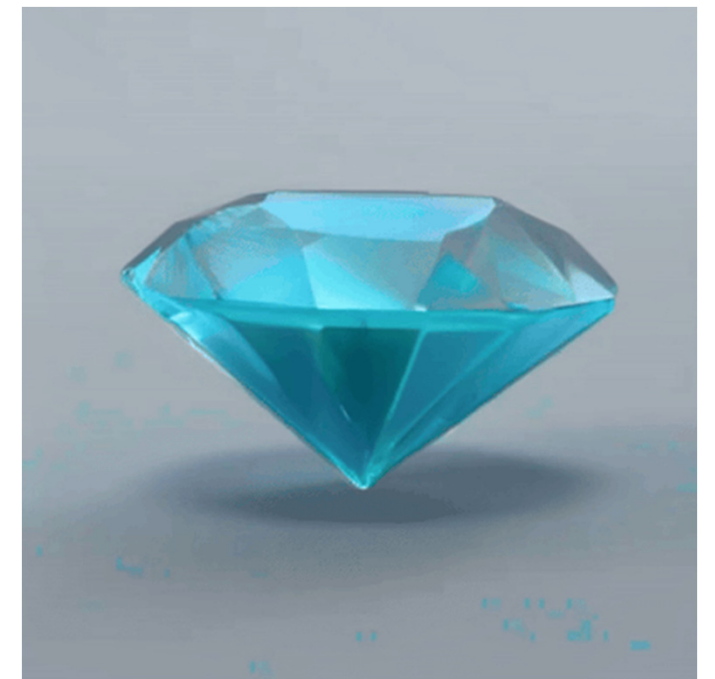
# Prior Work



PhysGaussian



PhysDreamer



PhysMotion

PhysGaussian: Physics-integrated 3D Gaussians for Generative Dynamics. CVPR 2024.

PhysDreamer: Physics-Based Interaction with 3D Objects via Video Generation. ECCV 2024.

PhysMotion: Physics-Grounded Dynamics From a Single Image. arXiv 2024.

# Prior Work

Physical dynamics are typically **added after**  
**shape reconstruction/generation**

PhysGaussian

PhysDreamer

PhysMotion

PhysGaussian: Physics-integrated 3D Gaussians for Generative Dynamics. CVPR 2024.

PhysDreamer: Physics-Based Interaction with 3D Objects via Video Generation. ECCV 2024.

PhysMotion: Physics-Grounded Dynamics From a Single Image. arXiv 2024.

# Parallel work



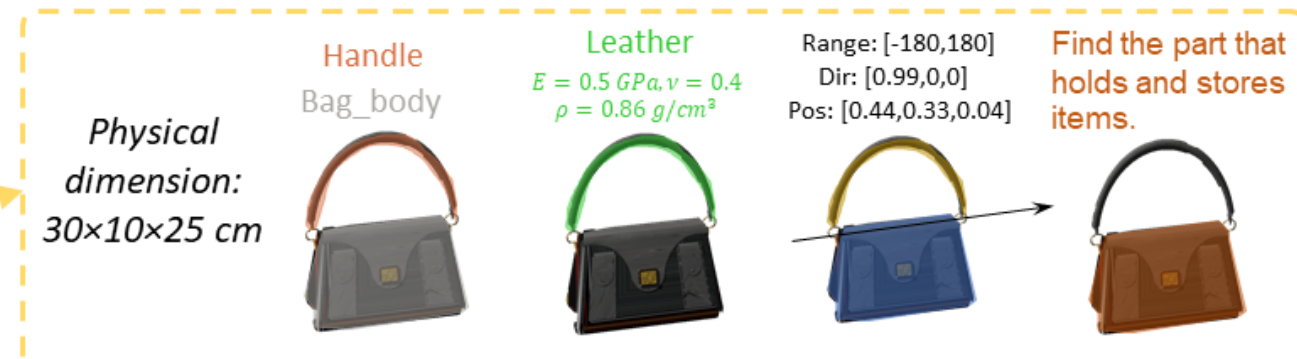
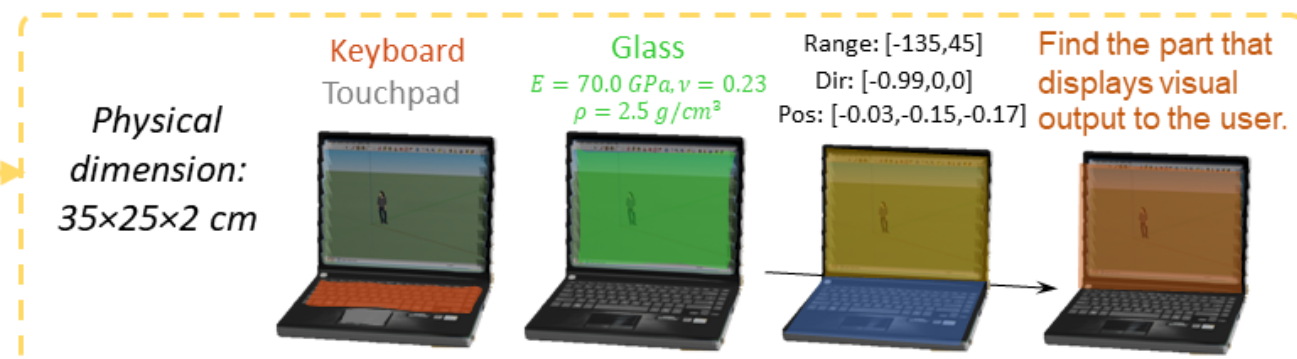
Officer Chair



Laptop

Handbag

## Physical properties



# We need datasets that include detailed material parameters!

## Material Parameters

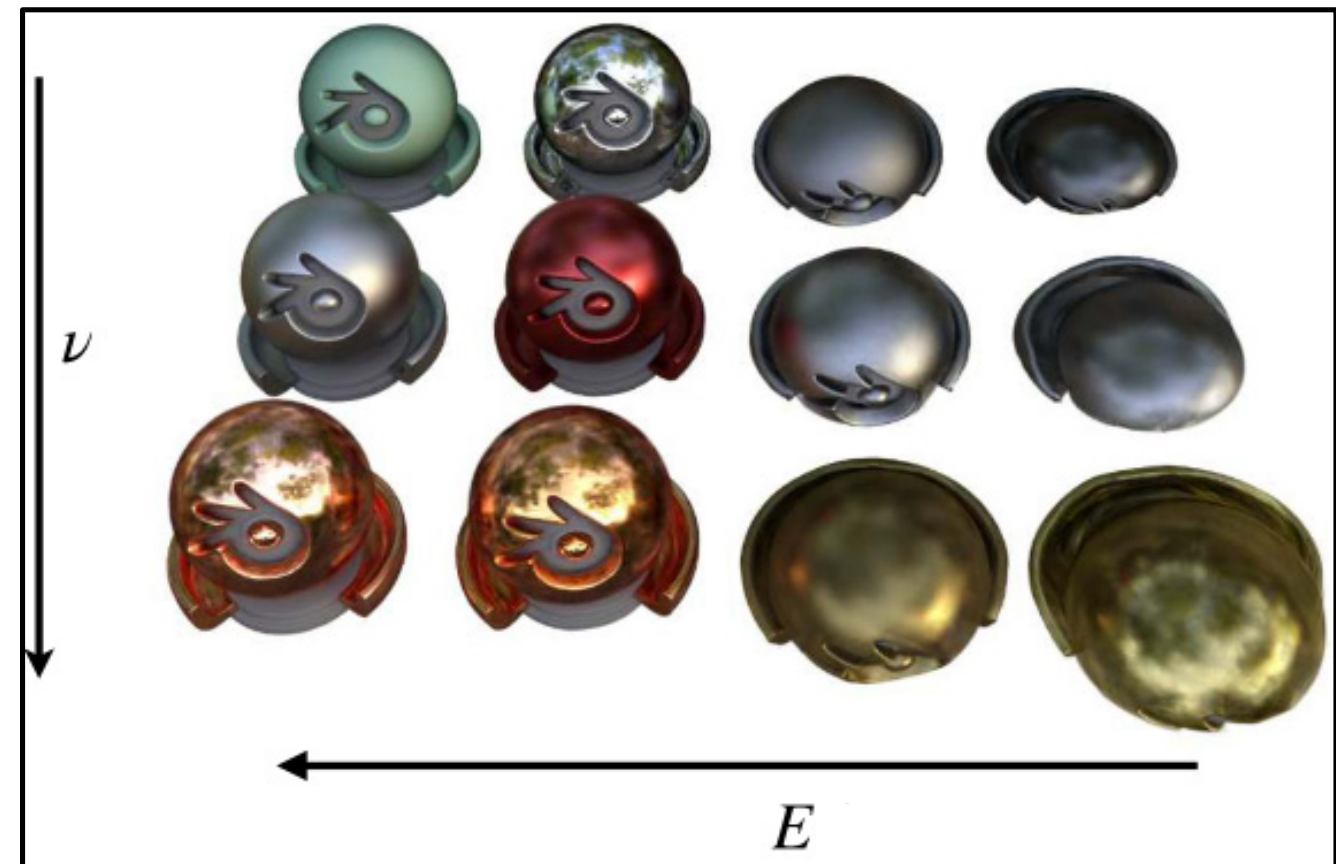
Young's modulus  $E$

Poisson's ratio  $\nu$

Yield stress  $\sigma$

Friction angle  $\phi$

...



# Material-annotated Dataset

## Assistance from Vision Language Models (VLMs)

### Rendered Images



### Text Descriptions

A chair made of 3 parts:  
(1) a pewter gray **metal**  
*leg*, (2) a dark navy  
blue **fabric** *seat*, and (3)  
a blue, gray stripes  
**fabric** *backrest*.

### Query Prompts

Please choose the most  
suitable **fine-grained**  
**material type** for the  
part(s): [seat,  
*backrest*] . . .  
Available choices for  
**fabric**: [cotton,  
wool, . . .] . . .

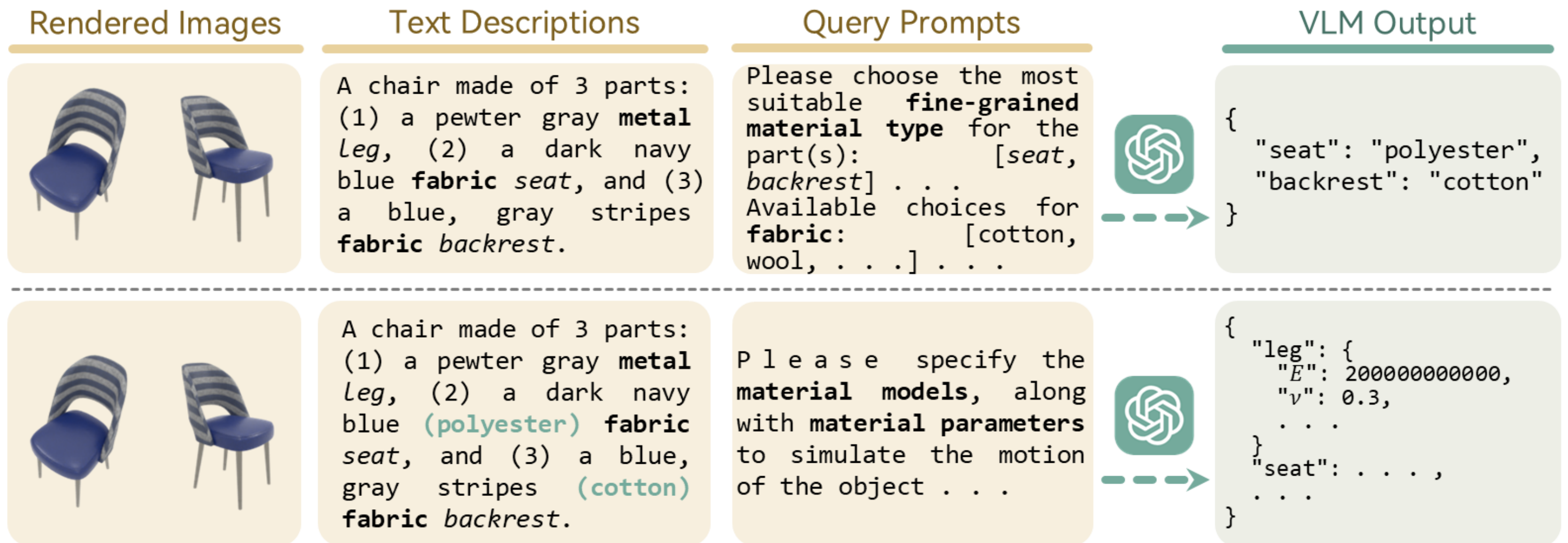


### VLM Output

```
{  
  "seat": "polyester",  
  "backrest": "cotton"  
}
```

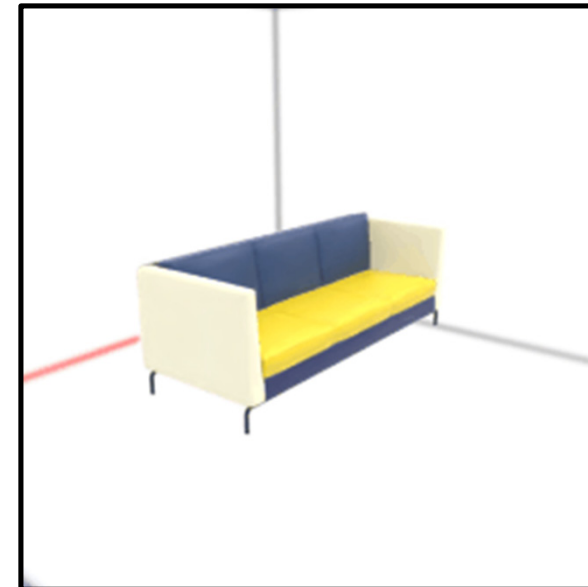
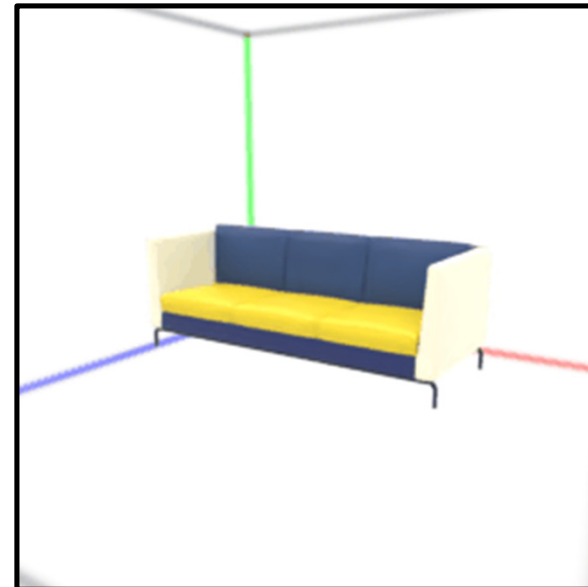
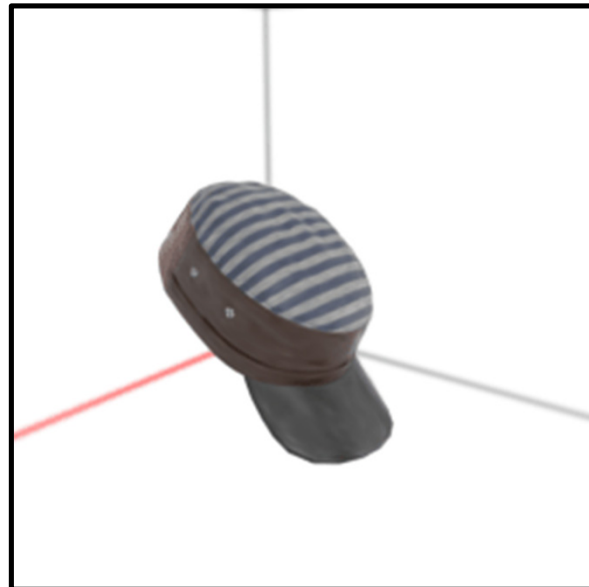
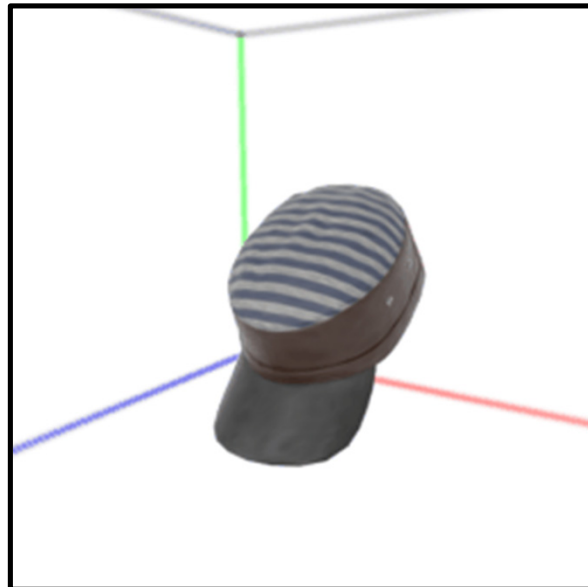
# Material-annotated Dataset

## Assistance from Vision Language Models (VLMs)



# Simulations

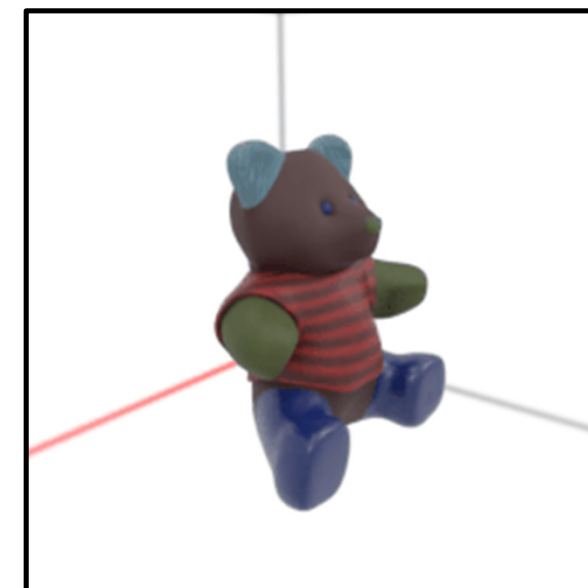
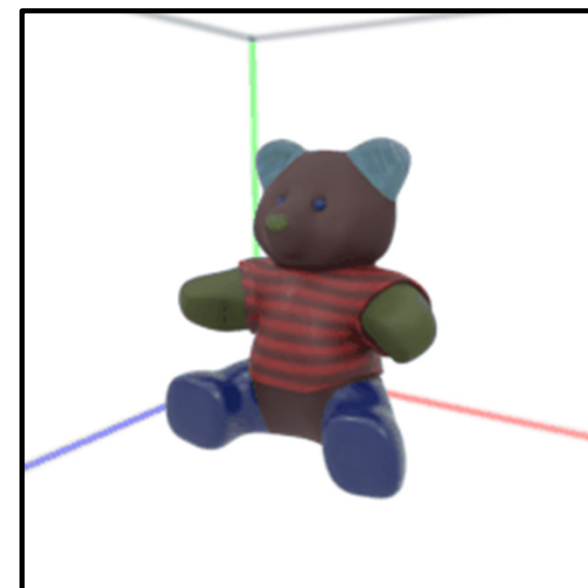
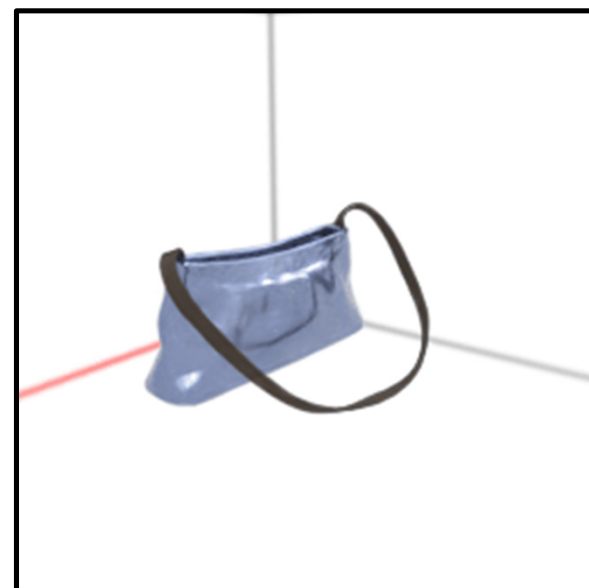
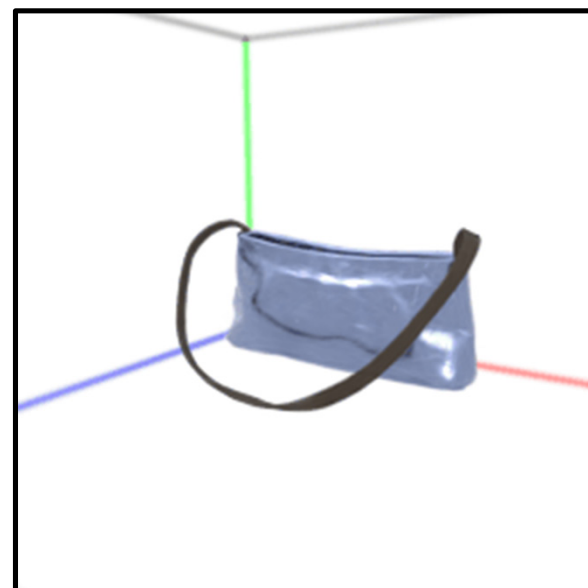
Drop



**Hat**

**Sofa**

Throw

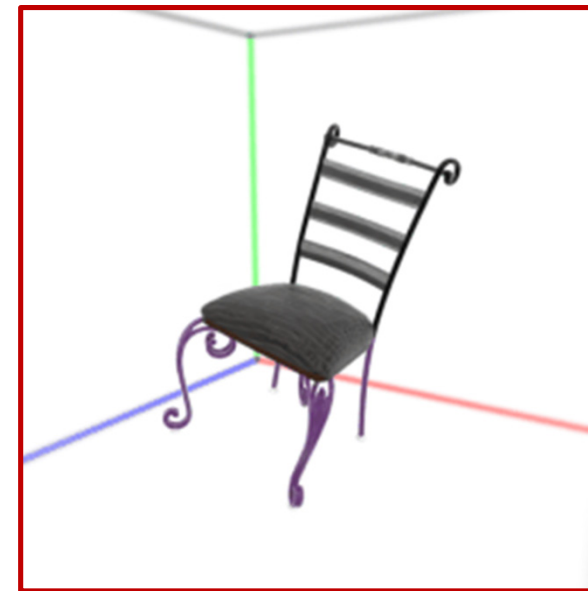
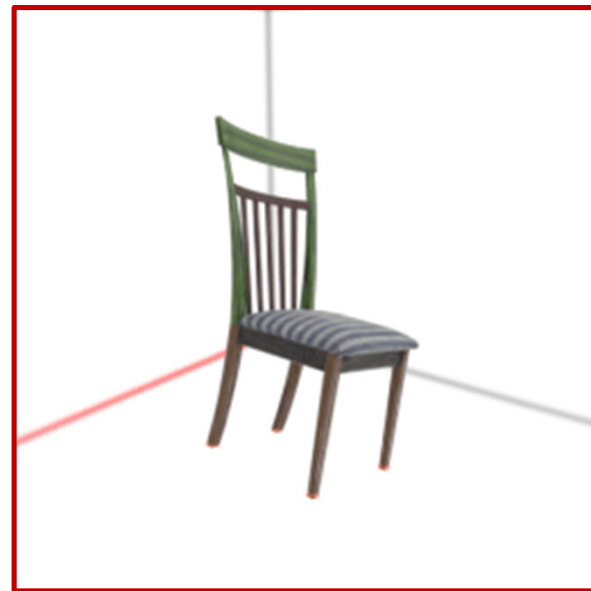
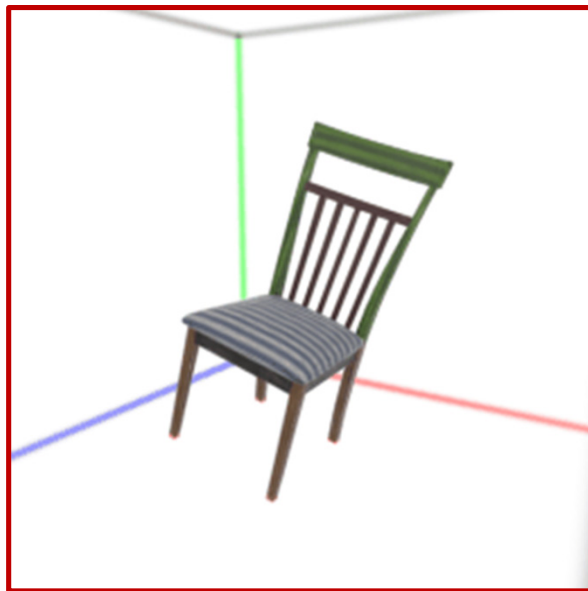


**Bag**

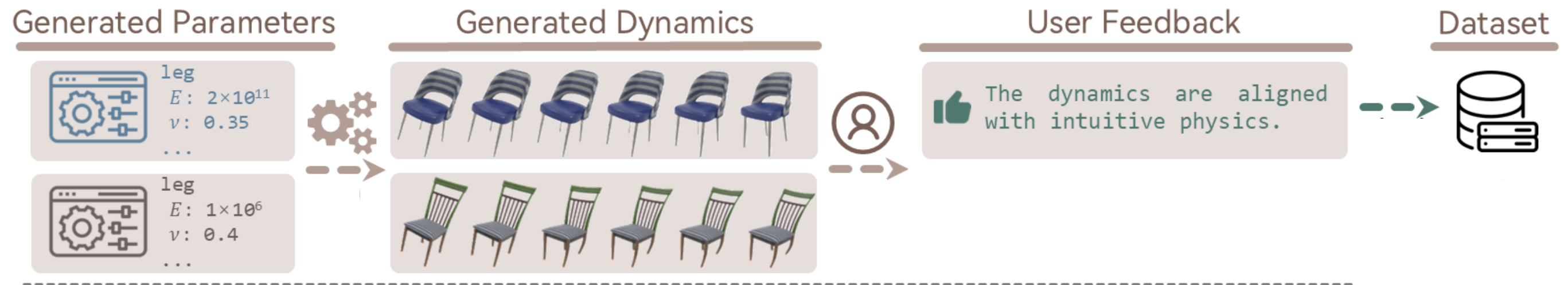
**Teddy Bear**

# Material-annotated Dataset

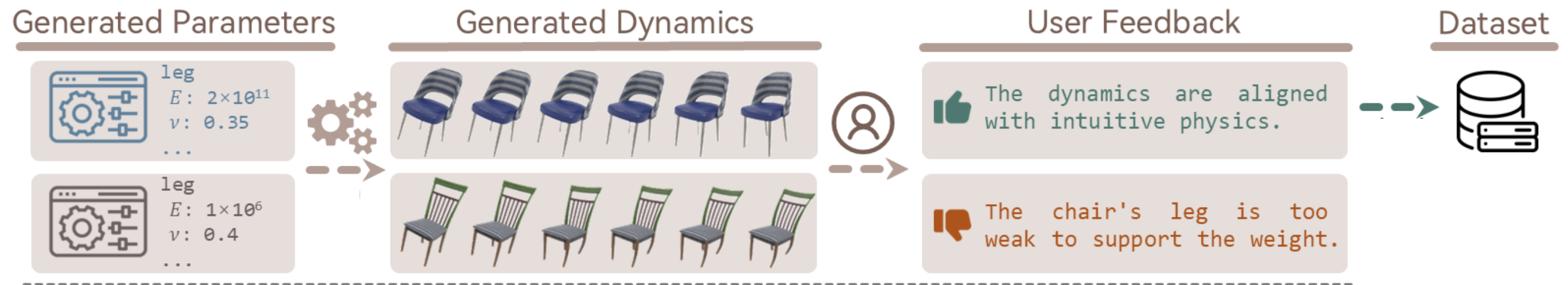
**Expert verification** for checking simulation results from the VLM parameters!



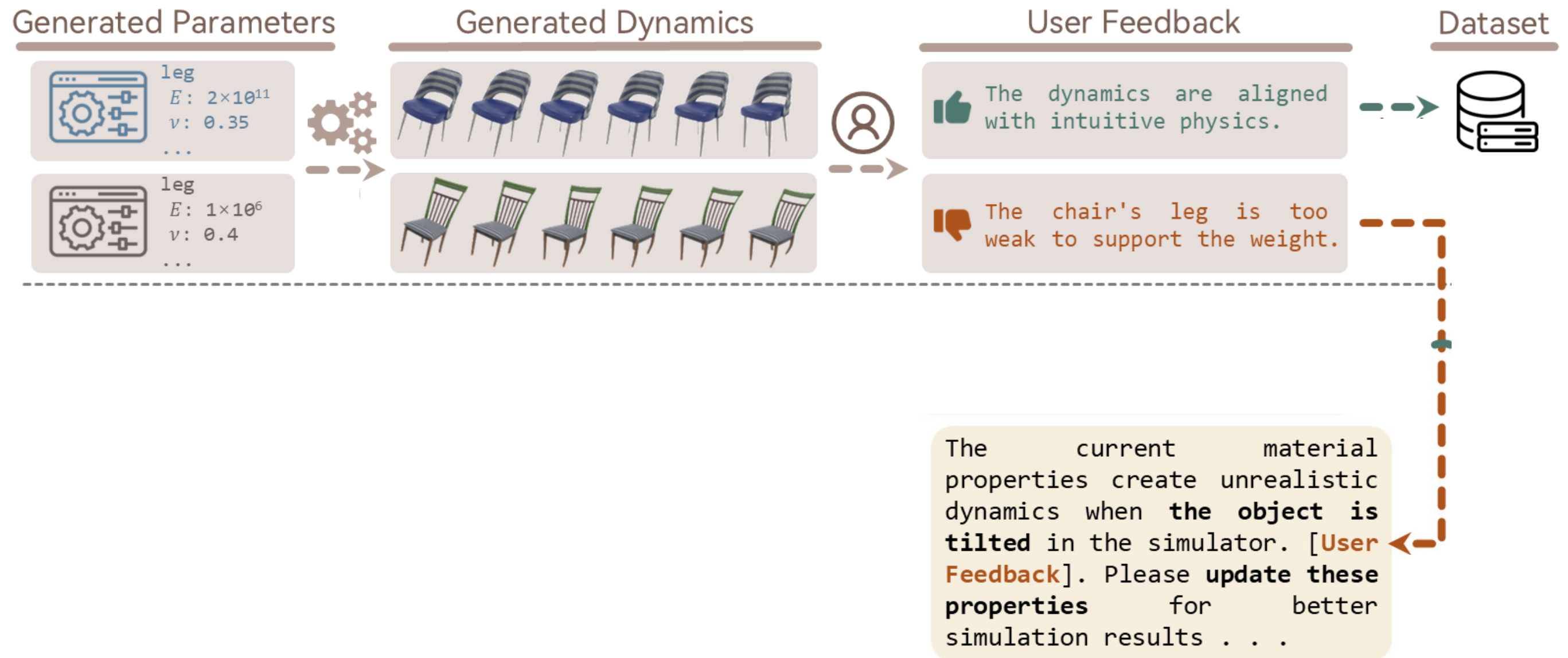
# Annotation Pipeline



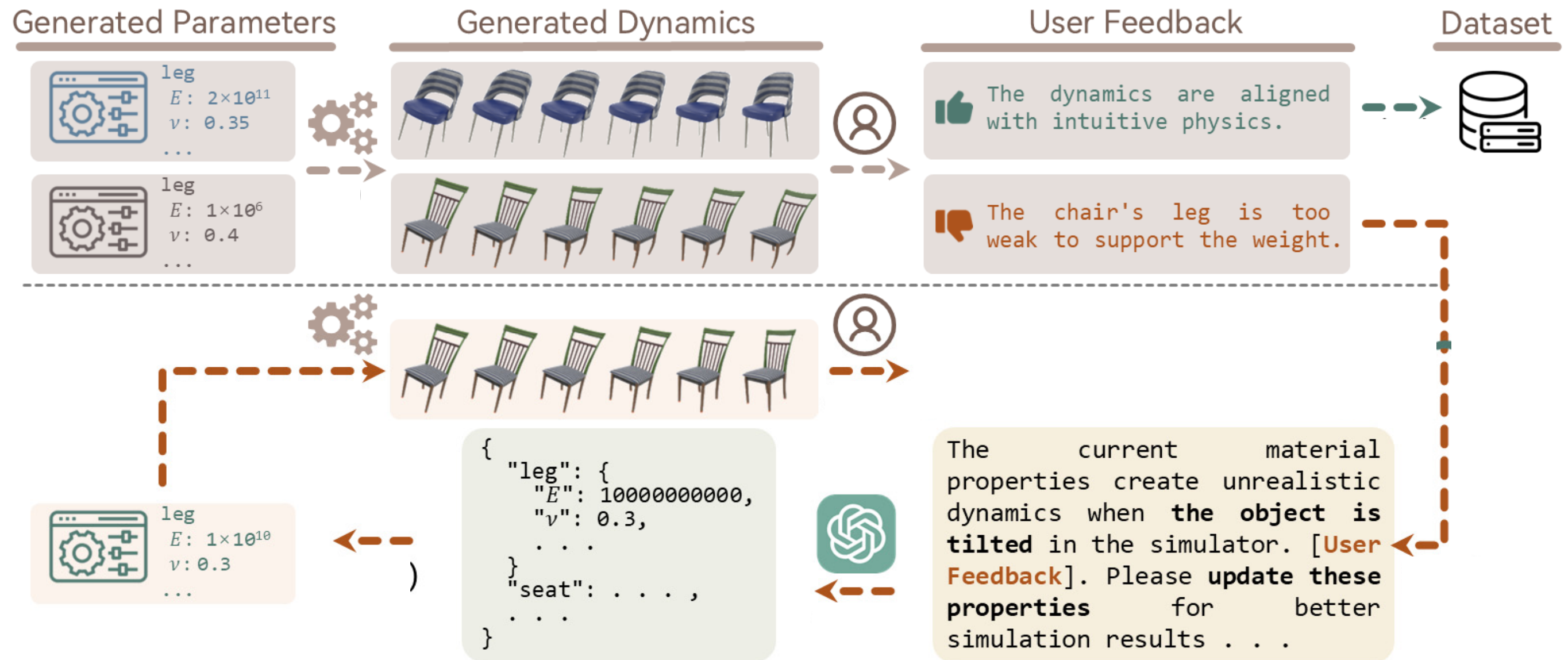
# Annotation Pipeline



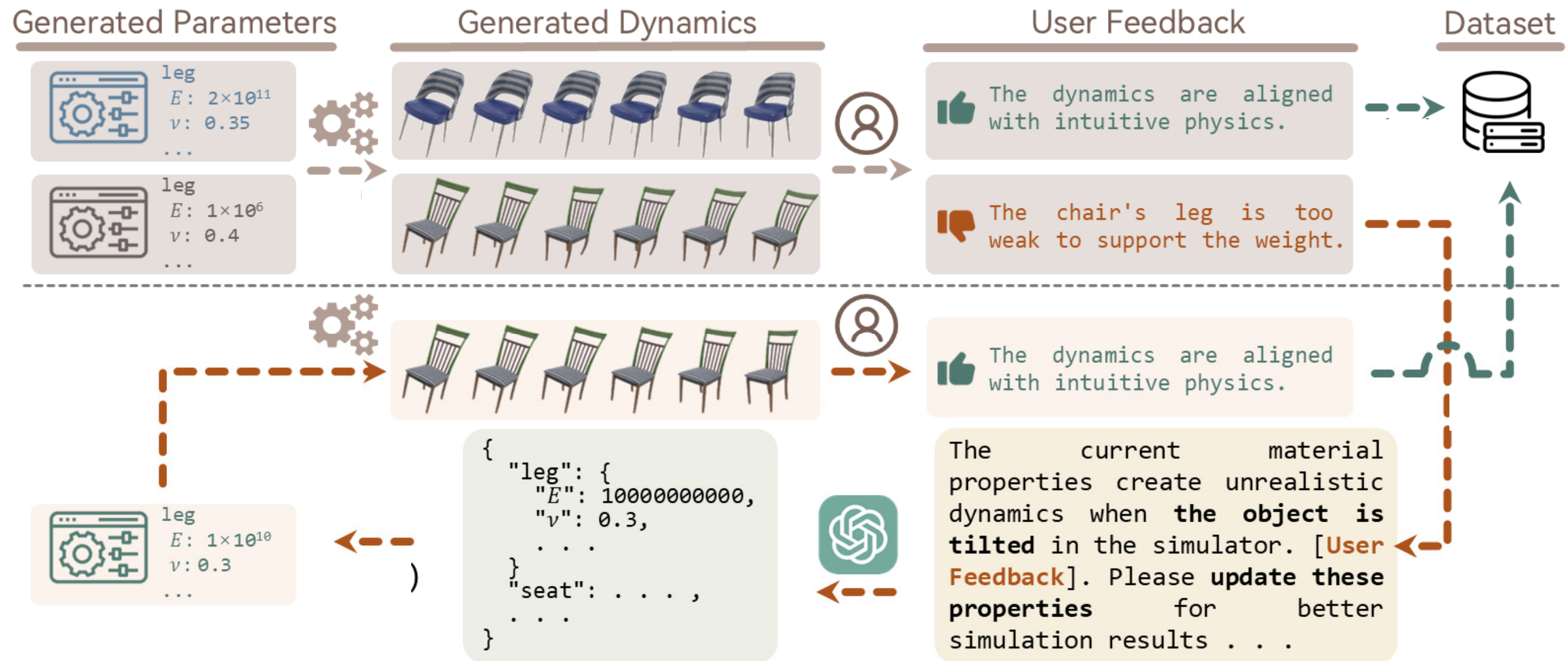
# Annotation Pipeline



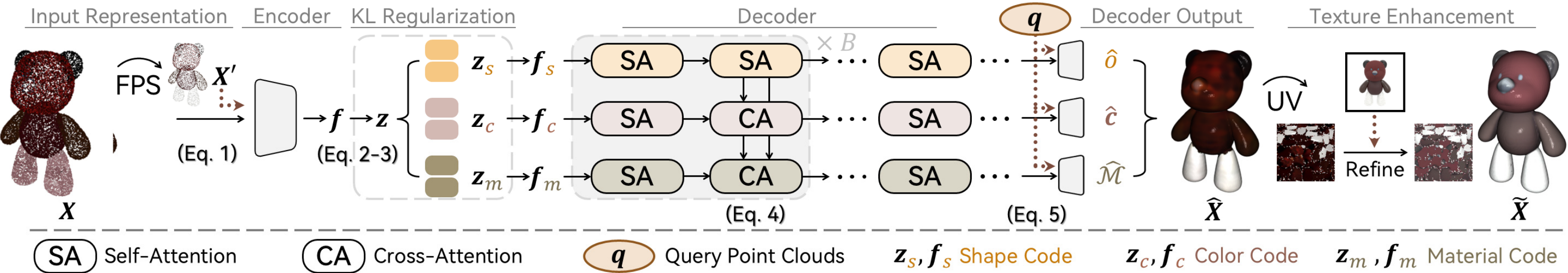
# Annotation Pipeline



# Annotation Pipeline



# Learning latent spaces of geometry+texture+materials



# Input Representation



For each surface point  $p \in P...$

{

position,

rgb,

Young's modulus,

Poisson's ratio,

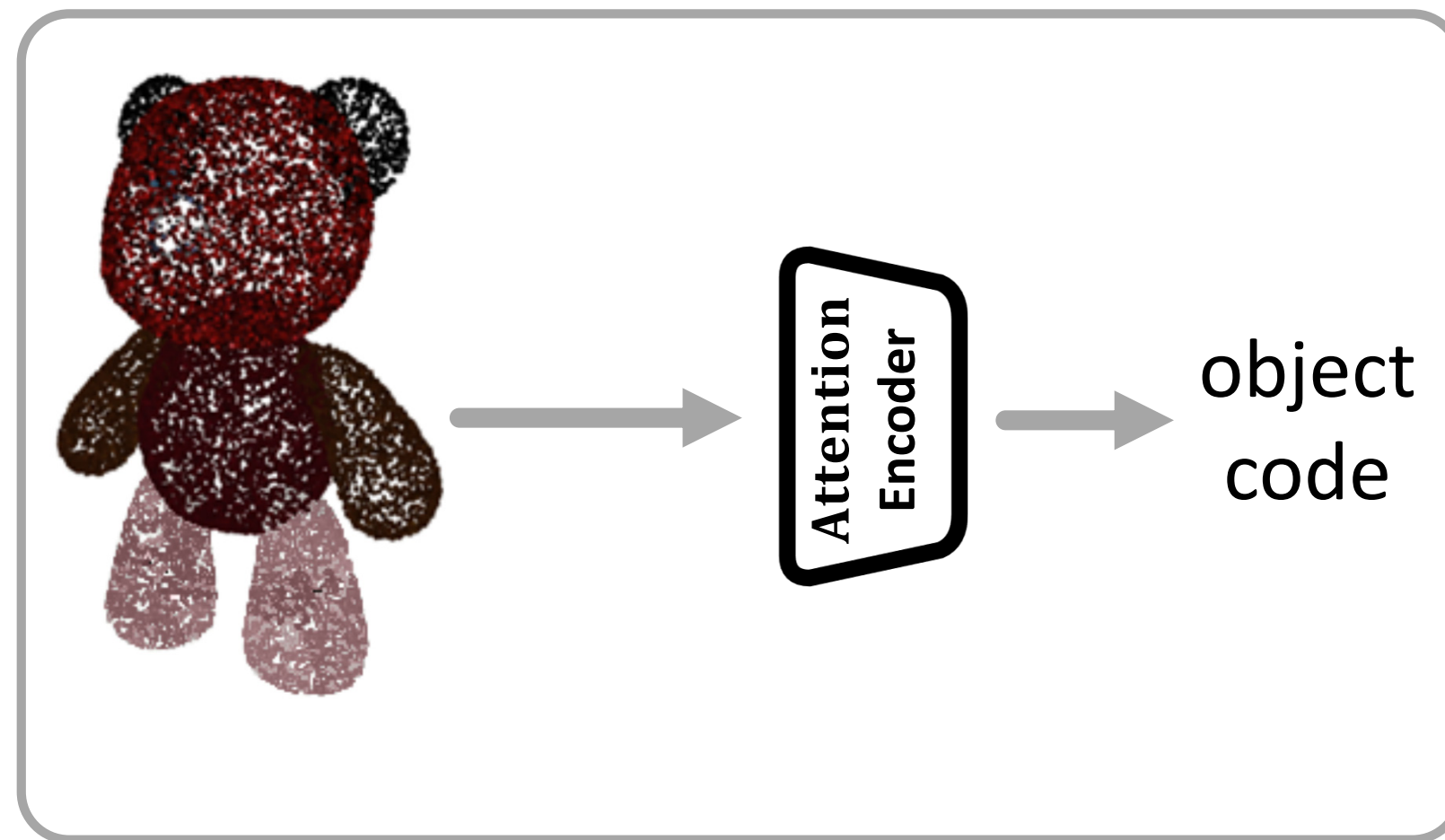
yield stress,

friction angle,

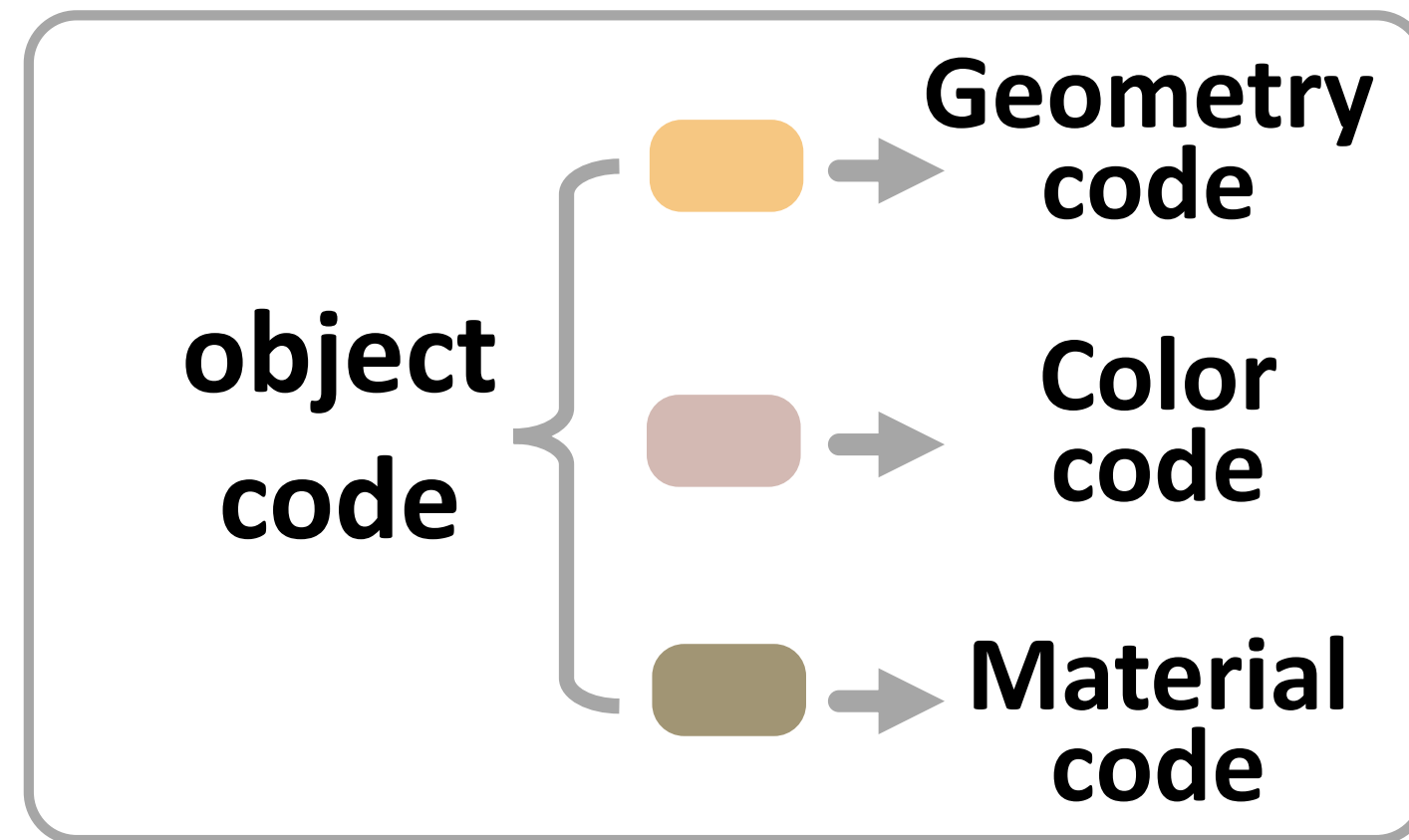
material type...

}

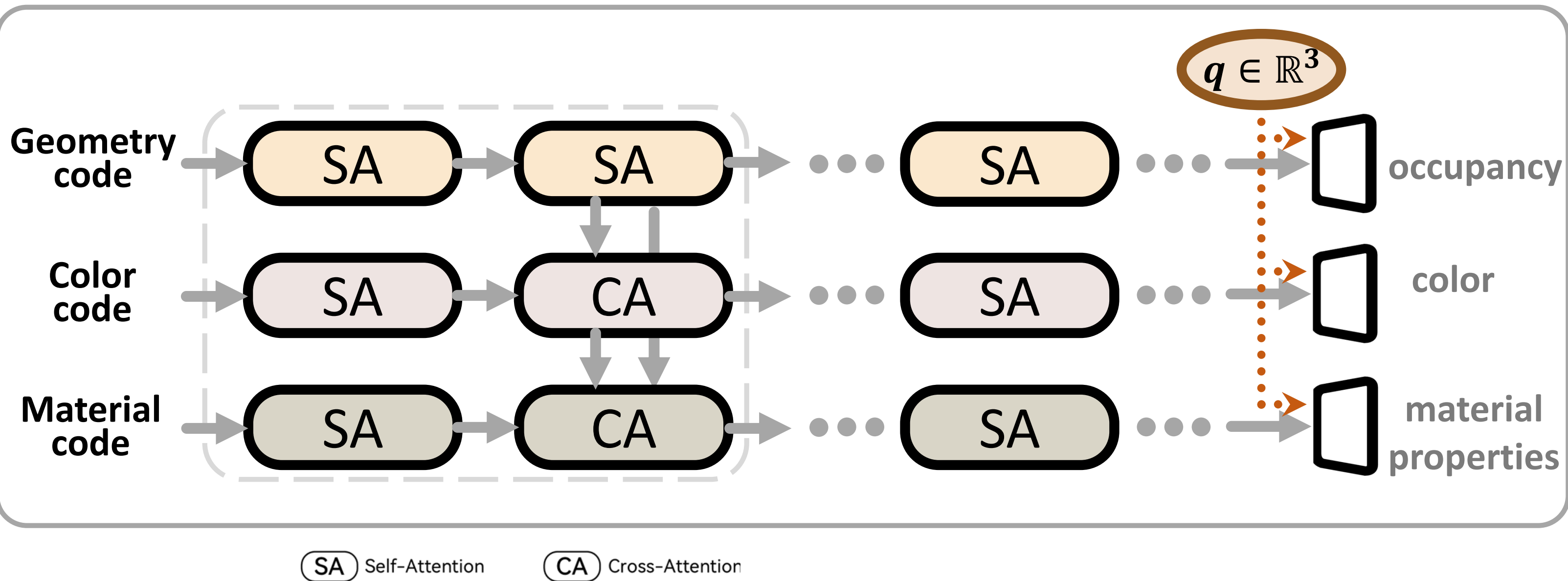
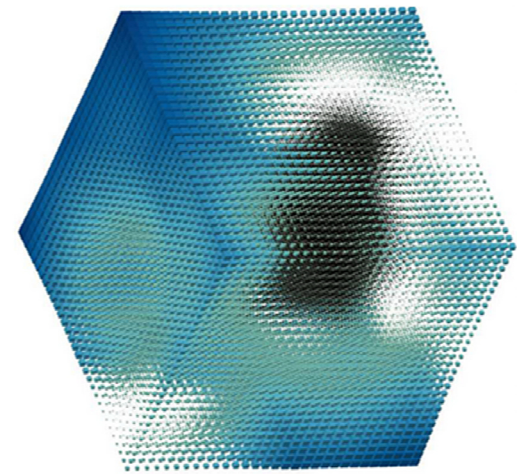
# Shape & Texture & Material Encoder



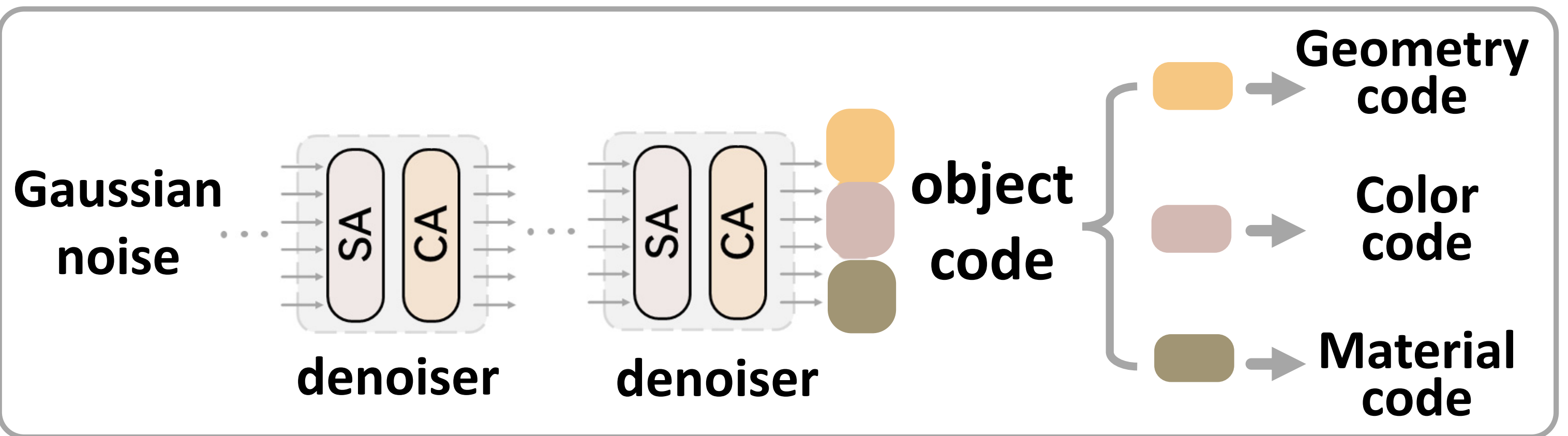
# Generative Model: code decomposition



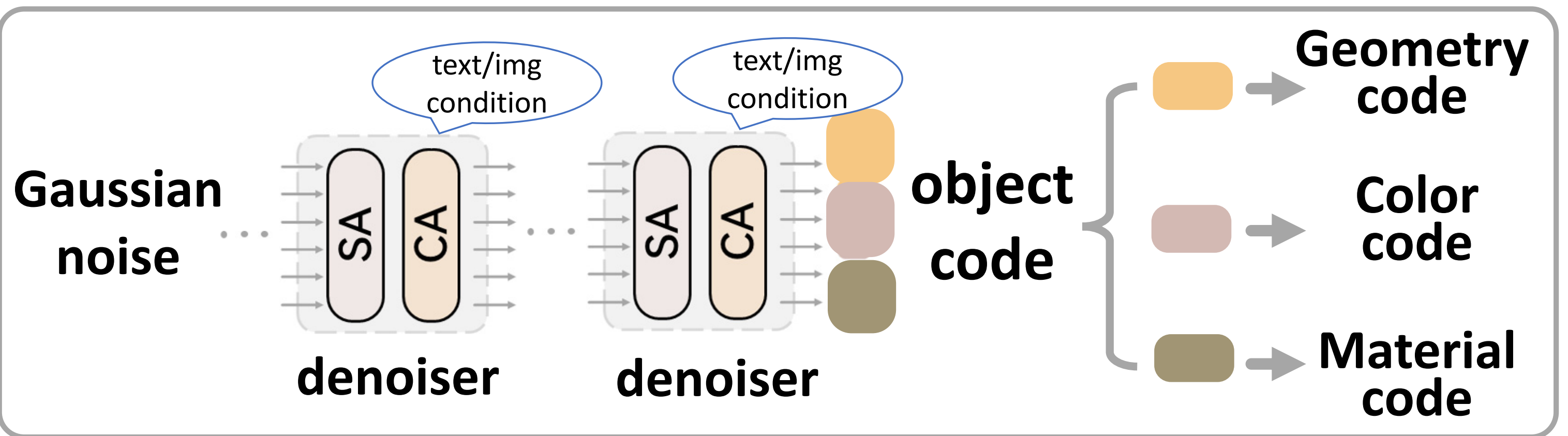
# Decoder



# Generative Model: diffusion



# Generative Model: diffusion



# Evaluate **physical commonsense**, not only renderings



Input image

Shape, then material decoder



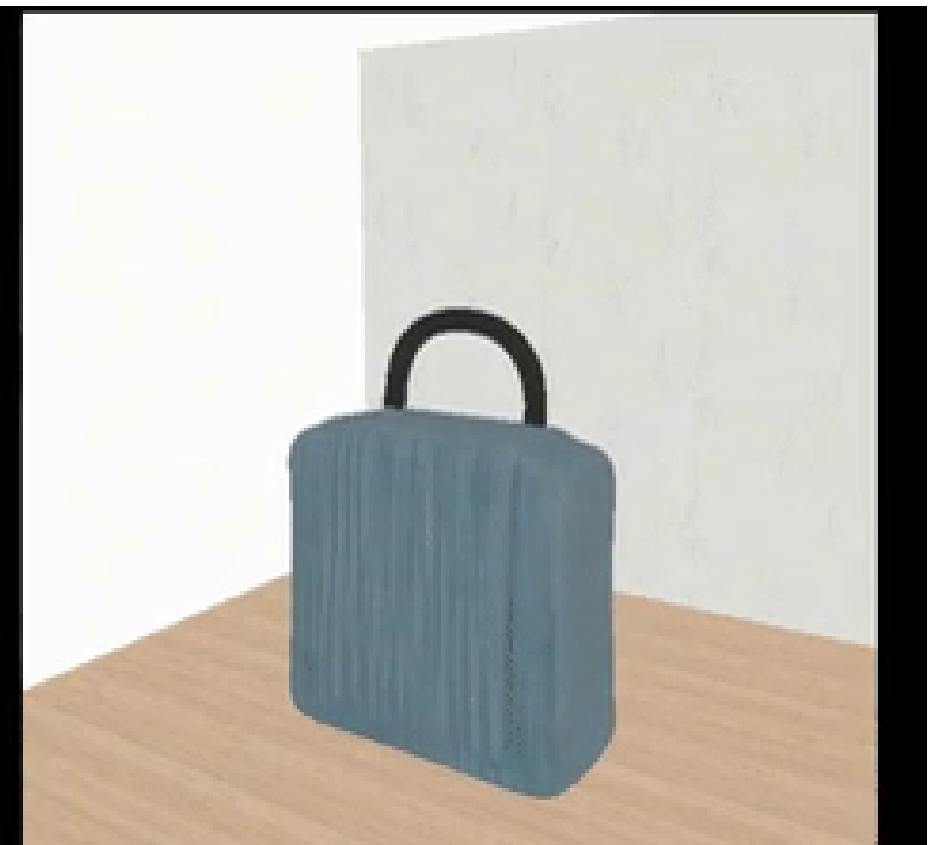
**-0.15** VideoPhy ↑

Shape, then material perception



**0.21** VideoPhy ↑

**SOPHY - Joint generation**



**0.43** VideoPhy ↑

# Roadmap

1

**GEOPARD**

What articulations  
are plausible for  
given assets?

2

**Rigel3D**

How to generate  
assets with their  
articulation?

3

**SOPHY**

How should assets  
respond in a  
physical world?

4

**BLARM**

How can we  
drive the motion  
of assets?

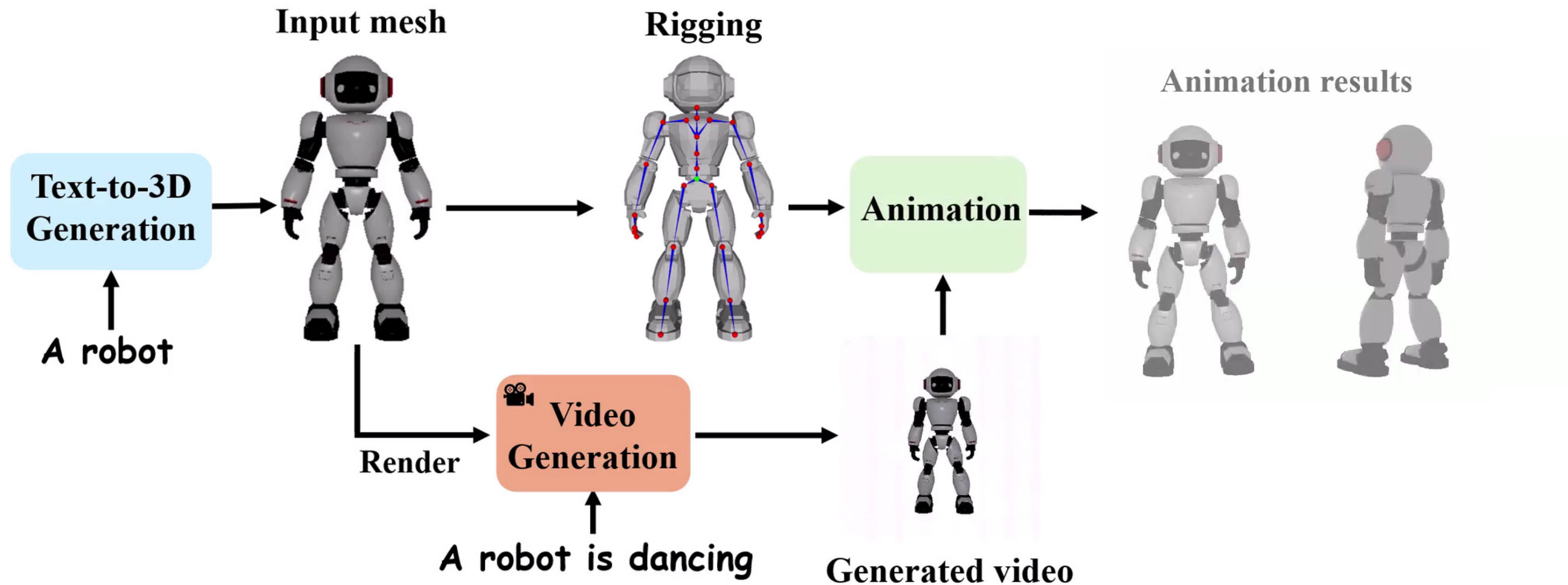


Pradyumn Goyal, Y.B. Shabat, H.T.D. Liu, H. Jiang, S. Mukherjee, K. Spence, M. Stauber, E. Kalogerakis, Y. Zeng

**BLARM:** Animating 3D Objects from Video via Blending LAtent Rigid Motion Primitives

<https://arxiv.org/html/2608.31113v1>

Prior work: Rig-based animation: compact structure,  
but potentially motion-mismatched



Prior work: Direct vertex motion -- expressive, but high-dimensional

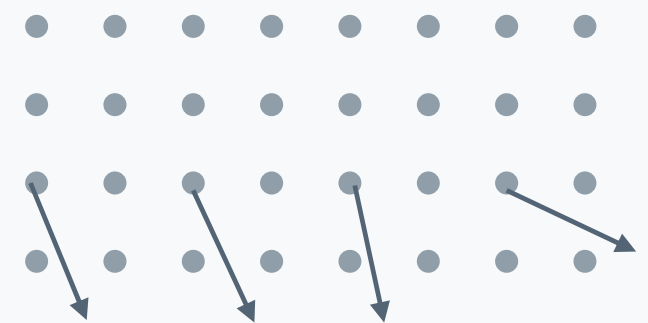
ActionMesh



Motion 3-to-4



Vertex trajectories



Dense deformation

# Can we get the best of both worlds?

## Two Different Animation Paradigms

### **EXPLICIT RIG**

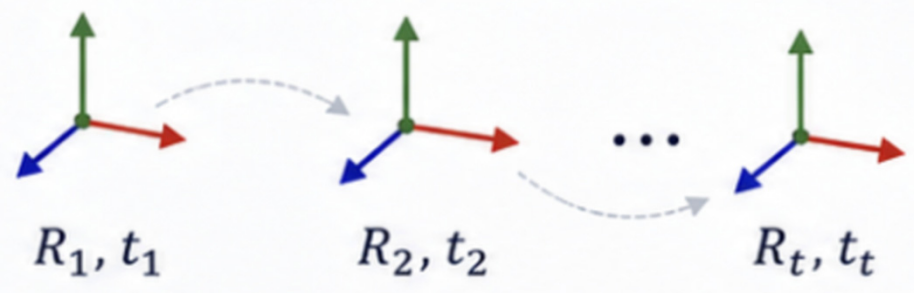
- Compact
- Structured
- Interpretable

### **DIRECT VERTEX MOTION**

- Expressive
- Motion-Adaptive
- Rig-Free

**Can we learn a compact & interpretable deformation space from geometry & observed motion without an explicit rig?**

# Key Idea: Learn latent rigid motion primitives from input video



**Rigid transforms vary with time**

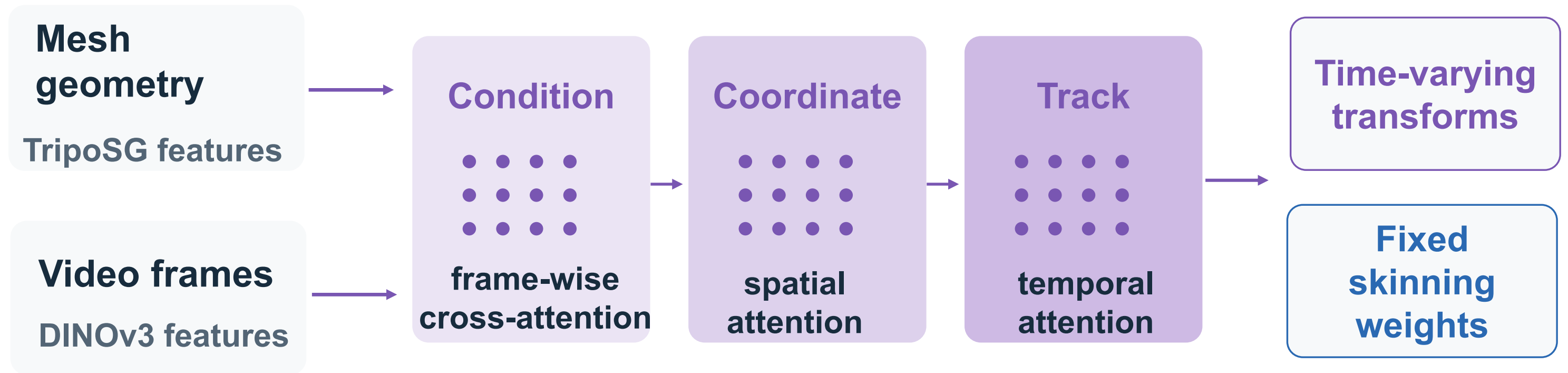


**Skinning weights stay fixed within the video sequence**



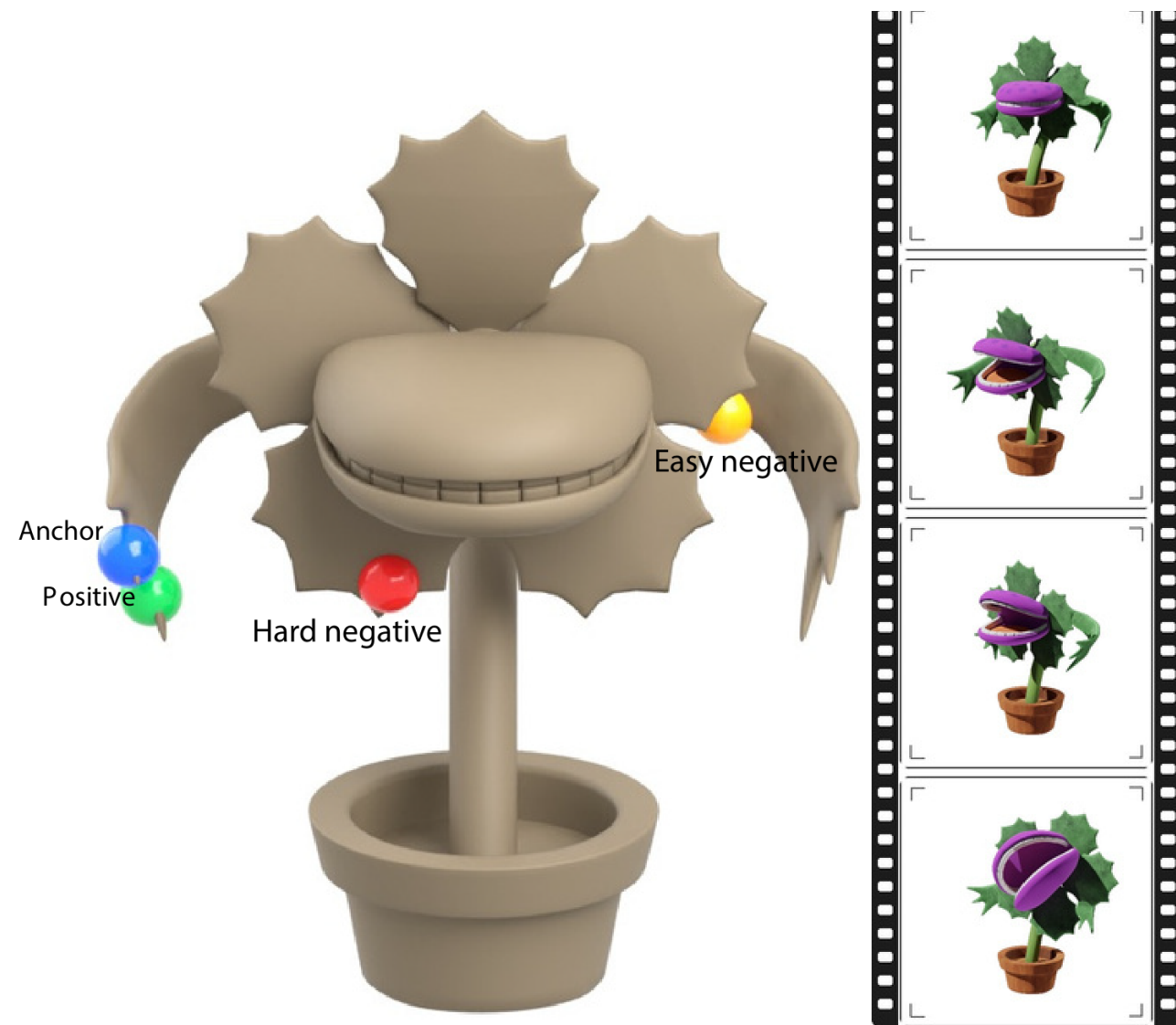
**Mesh Animation**

# Compact tokens link geometry and time



Factorized attention: coordinate motion components within frames, and each component across frames.

# Motion supervision organizes the latent space



## 1 Reconstruct trajectories

Match ground-truth 3D motion

## 2 Encourage sparse influence

Few components per vertex

## 3 Group consistent motion

Contrastive assignment learning

No explicit rig labels. Only ground-truth 3D trajectory supervision.

# Compact motion structure improves reconstruction

Mesh4D



BLARM



Reference



Motion correspondence error ↓

ActionBench

**9.24 → 7.15**

22.6% below the best compared baseline

Motion80

**12.78 → 8.72**

31.8% below the best compared baseline

A shared motion structure preserves shape while coordinating deformation over time.

## More qualitative comparisons

Action Mesh



Motion 3-to-4



Mesh4D



BLARM



Ground Truth



# Four representations, four different promises

| WORK    | REPRESENTATION         | WHAT IT ENABLES        |
|---------|------------------------|------------------------|
| GEOPARD | Part motion parameters | Plausible articulation |

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# Four representations, four different promises

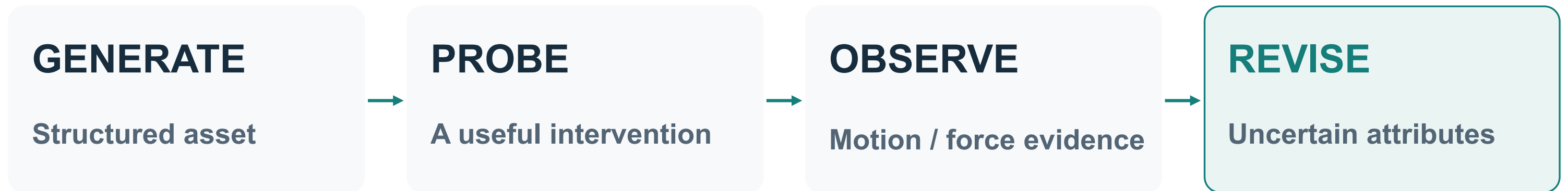
| WORK    | REPRESENTATION                     | WHAT IT ENABLES         |
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| Rigel3D | Skeleton + skinning                | Reusable pose controls  |
| SOPHY   | Physical material fields           | Simulated response      |
| BLARM   | Latent rigid motion + skin weights | Observed mesh animation |

# Four representations, four different promises

| WORK    | REPRESENTATION                     | WHAT IT ENABLES         | NOT YET ESTABLISHED         |
|---------|------------------------------------|-------------------------|-----------------------------|
| GEOPARD | Part motion parameters             | Plausible articulation  | Complete physical mechanism |
| Rigel3D | Skeleton + skinning                | Reusable pose controls  | Force-conditioned dynamics  |
| SOPHY   | Physical material fields           | Simulated response      | Calibrated real-world twin  |
| BLARM   | Latent rigid motion + skin weights | Observed mesh animation | Counterfactual interaction  |

The common principle: encode the structure behind motion and response.

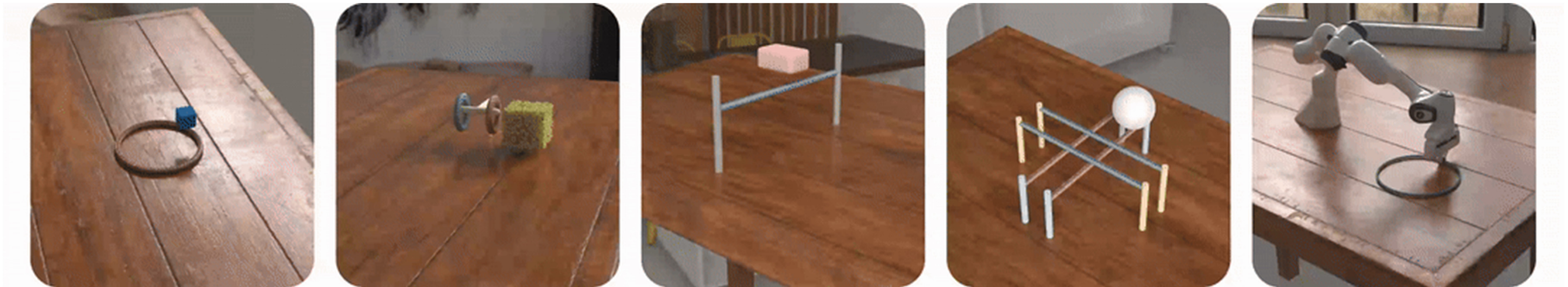
# From a recovered trajectory to an interactive model



**Make interaction a source of supervision, not only a downstream application.**

# From interactive assets to interactive worlds

- Generative models of **object geometry + articulation + materials + motions**
- **4D scene generation** with multiple **objects + agents** interacting



# Thank you!

## *Student Team:*



*Dmitry Petrov  
PhD, UMass,  
(now @ Foundation)*



*Junyi Cao  
PhD  
UMass & TU Crete*



*Vikas Thamizharasan  
PhD  
UMass*



*Pradyumn Goyal  
PhD  
UMass*



*Tuan Ngo  
PhD  
UMass*



*Nikitas Chatzis  
MS  
TU Crete & NTUA*



*Marios Loizou  
Postdoc  
TU Crete & CYENS*

*Funding:*



*Project pages &  
Links to students' pages:*

<https://kalo-ai.github.io/>

